

**THE IMPACT OF GOVERNMENT
AGROMERCADOS ON PRICE DISPERSION
AND FOOD SECURITY**

*Evidence from a Natural Experiment in El
Salvador*

By

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Vienna, 08 June 2026

Abstract

I assess whether El Salvador's Agromercados programme reduced private food prices and price dispersion in nearby district markets. The analysis uses a high-frequency bi-weekly panel of 43 district markets, 12 vegetables, and 95 periods, combined with a doubly robust staggered difference-in-differences design that exploits staggered rollout and an administrative blackout of government price broadcasts. I find no statistically significant average treatment effect on either log private prices or basket-level price dispersion, implying that physical entry alone did not generate market-wide price convergence.

Further evidence points to a limited mechanism. Active broadcasting is associated with lower price dispersion but not lower private prices, while the private-public price gap widens over time and pass-through, or price elasticities are asymmetric. Competitive pressure is also spatially concentrated near agromercados. The programme therefore appears to have functioned as a localised and incomplete price anchor, rather than as a binding substitute for private market power.

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Chapter 1

Introduction

When a government enters a retail market as a highly visible, subsidised competitor, standard mixed-oligopoly theory predicts that private prices should converge towards the state's benchmark ([Merrill and Schneider, 1966](#); [Obradovits and Plaickner, 2023](#)). But what happens when structural market frictions—such as physical distance and consumer search costs, insulate private vendors from public competition ([Hotelling, 1929](#); [Stigler, 1961](#))?

In mid-2024, El Salvador experienced a severe episode of food price inflation. Triggered initially by climate-driven supply disruptions, the crisis quickly revealed deeper structural vulnerabilities in the domestic food supply chain. The Ministerio de Agricultura y Ganadería (Ministry of Agriculture and Livestock; MAG) and the government publicly attributed the magnitude and persistence of retail price spikes not solely to weather shocks, but to possible rent extraction by wholesale intermediaries leveraging the crisis to widen their mark-ups ([BBC News Mundo, 2024](#)).

Retail food markets in El Salvador are structured in ways that make this kind of extraction feasible. District markets operate as one of the primary retail channels for fresh produce, and are characterised by multiple small vendors who price loose produce independently, without posted ceilings. Price comparison requires a physical visit to each stall, and vendor–buyer relationships are sustained by informal credit and geographic proximity. These conditions generate the classical prerequisites for equilibrium price dispersion above the competitive level ([Stigler, 1961](#)): rational mark-ups sustained by search frictions and captive consumer bases, rather than genuine product differentiation.

The policy response and natural experiment. The government's response was an immediate and aggressive attempt to bypass this intermediary market power. On 6 July 2024, MAG launched the *Agromercados* programme: new government-operated retail market places sourcing produce directly from cooperatives and selling at a centrally set, highly subsidised weekly price, P_t^{gov} . The programme expansion to over sixty sites by early 2025, suggests new locations rolling out in a staggered sequence determined by logistical readiness rather than local price conditions ([Legislative Assembly of El Salvador, 2025](#)) ([elsalvador.com, 2024](#)). This staggered

rollout provides the first treatment indicator and source of quasi-experimental variation I leverage in my study.

Crucially, the intervention was not merely physical; it was also mediated by information shock. Once *Agromercados* were introduced, MAG massively broadcasted the weekly P_t^{gov} across national media channels, collapsing marginal consumer search costs and force the market into a price-directed search regime (Ding and Zhang, 2018).

The introduction of the agromercado programme should alter this equilibrium in two ways. First, a new low-price public firm disrupts the market and should incentivise competition amongst private vendors (district markets). Secondly, by broadcasting weekly agromercado prices P_t^{gov} through official government channels, the programme breaks down information barriers and reduces consumer’s search costs.

These two mechanisms—physical competition and information signalling, have not been simultaneously active throughout the intervention. On 28 November 2024, responding to Central Bank concerns about deflationary pressures, MAG abruptly ceased national broadcasting (Diario El Mundo, 2024). Physical markets remained open, but the information channel was switched off for the subsequent ten months ($\text{Info}_t = 0$). This administrative blackout, driven by macroeconomic rather than local market considerations, provides my second treatment indicator and an opportunity to separate the physical causal effect of a state competitor from the effect of a price information signal to consumers and producers alike.

In this study, I examine the causal impact of the Agromercados programme on private-market prices in surrounding district markets. The central question is:

Did the introduction of government-subsidised agromercados reduce private food prices and price dispersion in nearby district markets, and through which channel?

This is addressed along three dimensions:

- **The price-anchoring channel:** Did the physical entry of an agromercado force private prices downward, and did active price broadcasting amplify this effect?
- **The food security channel:** Did the programme reduce within-district price variation, acting as a consumption-smoothing mechanism for local households (Bellemare, 2015)?
- **The spatial channel:** How rapidly does the disciplining effect decay with road-network distance, and at what boundary does private market power reassert itself?

Main Findings Using a high-frequency biweekly panel of twelve basic vegetables across 43 of El Salvador’s district markets, and a doubly-robust staggered difference-in-differences estimator (Callaway and Sant’Anna, 2021), the central result is an empirically estimated null. Agromercado entry did not generate market-wide price convergence: the aggregate average

treatment effects on both log prices and basket price dispersion are statistically indistinguishable from zero. Physical market presence alone was insufficient to discipline private vendors.

Beyond this aggregate null, however, the programme operated as a *localised and incomplete price anchor*. Three findings support this characterisation. First, active information broadcasting independently compressed within-district price dispersion, but did not reduce the level of private prices—vendors moved closer to one another without moving closer to the public benchmark. Second, the private–public price gap widened progressively with relative exposure time to an agromercado, and pass-through was sharply asymmetric: private prices tracked P_t^{gov} upwards but were unresponsive to downward adjustments, indicating that vendors used the public reference as a ceiling rather than a constraint. Third, any competitive discipline was tightly localised. Together, these results indicate a market in which the state coordinated private pricing behaviour more than it disciplined it.

Contributions and Structure This thesis makes three primary contributions:

- **Empirically**, it provides the first causal evaluation of El Salvador’s Agromercados programme using a staggered difference-in-differences framework robust to heterogeneous treatment timing, applied to a panel spanning 43 districts and 95 periods. The precisely estimated null is itself informative: state retail entry is insufficient to discipline private vendors when structural travel and search frictions persist.
- **Theoretically**, the intervention’s mechanisms are decomposed by exploiting the administrative broadcasting blackout as a natural experiment. Utilising a sequential meta-regression, the estimated asymmetric pass-through elasticities ($\hat{\beta}^+ = 0.53$, $\hat{\beta}^- \approx 0$) prove inconsistent with price-directed search, indicating instead that informal vendors exercised residual local market power by treating the state price as a reference ceiling rather than a binding outside option (Obradovits and Plaickner, 2023).
- **In terms of policy relevance**, this study measures the spatial decay of competitive discipline against road-network distance. Furthermore, it leverages the coefficient of variation of a food basket as a reduced-form proxy for food security to quantify the programme’s impact on household price stability.

The remainder of the thesis is structured as follows. Chapter 2 reviews the theoretical literature on search frictions, mixed oligopoly, and spatial competition. Chapter 3 details the institutional context and programme timeline. Chapter 4 presents the data construction and treatment assignment. Chapter 5 outlines the empirical strategy. Chapter ?? presents the results and mechanism tests. Lastly, Chapter ?? summarises the findings and discusses policy implications.

Chapter 2

Literature Review

Price formation in El Salvador's food market combines fragmented private retail competition in district markets, high consumer search costs, and limited outside options for low-income households. To better understand how Agromercados altered this equilibrium, this analysis is built upon three pillars of theory: (1) search and information economics; (2) mixed oligopoly with a welfare-maximising public firm, and; (3) spatial competition.

2.1 Search Costs, Information and Equilibrium Price Dispersion

[Stigler \(1961\)](#) provides the analytical starting point, establishing that price dispersion is not a market failure but a rational outcome of costly consumer search. In Salvadoran district food markets—characterised by perishable goods, multiple vendors, and high information costs, this dispersion is a *de facto* market feature, allowing sellers to obtain mark-ups above marginal cost.

From random search to price-directed search: [Stahl \(1989\)](#), [Salop and Stiglitz \(1977\)](#), and [Varian \(1980\)](#) formalise how dispersion persists when consumers differ in search intensity: sellers price high to extract surplus from non-searchers and low to compete for comparison shoppers, sustaining a mixed-strategy equilibrium above marginal cost. [Ding and Zhang \(2018\)](#) extend the framework to settings where prices are observable ex-ante, modelling search behavior where consumers evaluate products on a binary, all-or-nothing basis. Building on this, [Obradovits and Plaickner \(2023\)](#) introduce variable product differentiation to reveal a central pricing paradox: when consumers can see prices upfront but must still incur search costs to evaluate product fit, a reduction in search costs actually leads to higher market prices. Because lower search costs increase a consumer's willingness to bypass cheaper, partially-matching goods to sequentially inspect higher-priced alternatives for a perfect match, competitive pressure is organically relaxed, allowing firms to shift their pricing strategies stochastically upward and expand their mark-ups.

The Agromercados programme was designed to trigger precisely this mechanism by posting a publicly announced weekly price and broadcasting it nationally. The information blackout of November 2024 provides a natural counterfactual: physical outlets remained open but the price signal was withdrawn, allowing to test the magnitude of the competitive effect, if any, attributed to the information channel rather than physical agromercado presence alone.

2.2 Mixed Oligopoly and the State as a Maverick Firm

Merrill and Schneider (1966) establish that the introduction of a welfare-maximising public firm into a private oligopoly acts as a powerful disciplinary mechanism, forcing profit-maximising incumbents to structurally lower their prices to retain their remainder demand. The Agromercados programme operationalises this "maverick firm" dynamic within a spatial landscape. By offering unified and subsidised prices P_t^{gov} , the state creates a strict ceiling on what private vendors should charge for comparable produce. However, because consumers' must incur search and transport costs τd_i to access the agromercado, this pricing constraint decays over distance, and varies from consumer to consumer. Consequently, the state functions a spatially bounded maverick firm: its disciplining effect against private price mark-ups is strongest for consumers with low search costs and vendors in close proximity, but distance organically restores private market power.

2.3 Spatial Competition and the Radius of Relief

Hotelling (1929) shows that transport costs generate local market power by weakening consumers' willingness to switch to more distant sellers. Even when products are otherwise similar, distance makes the outside option less binding and allows nearby vendors to preserve mark-ups.

Applied to Agromercados, this predicts that any competitive discipline imposed by a public outlet should attenuate with road-network distance. Private vendors located closer to an agromercado should face stronger competitive pressure than vendors serving consumers for whom travel is more costly.

The empirical implication is a spatial decay pattern in treatment effects rather than a single critical cut-off. Section 5.4 therefore tests whether estimated effects weaken across concentric distance bands around agromercado locations.

2.4 Price Volatility, Food Security and Welfare

The welfare argument of this study rests not only on price levels, but also on price stability. For low-income households, intertemporal price volatility matters because consumption smoothing

requires that basic food prices remain predictable from one period to the next. When households lack financial buffers, sudden price spikes can force them to reduce consumption, substitute towards lower-quality goods, or postpone purchases altogether.

Following [Bellemare \(2015\)](#), this thesis uses the coefficient of variation (CV) to capture local price instability while abstracting from differences in average price levels. In the empirical analysis, the CV_{it} is constructed within each district-period over the harmonised 12-product vegetable basket. This makes it a reduced-form proxy for the retail instability faced by consumers in local markets rather than a measure of aggregate inflation.

In principle, Agromercados could reduce this instability through two channels. First, the stable public benchmark price, P_t^{gov} , may limit private mark-up spikes. Second, when that benchmark is actively broadcast, unusually high private prices become more visible to consumers, raising the cost of sustaining temporary price surges. A fall in the within-district basket CV_{it} in treated areas is therefore interpreted as evidence that Agromercados functioned as local shock absorbers for households.

Chapter 5 operationalises this idea using the basket CV_{it} as the second outcome track within the staggered difference-in-differences framework.

2.5 Staggered Difference-in-Differences and Estimator Choice

The canonical two-way fixed-effects (TWFE) estimator is problematic in staggered-adoption settings. As [Goodman-Bacon \(2021\)](#) shows, TWFE implicitly combines multiple two-by-two comparisons, including already-treated units as controls for later-treated units. When treatment effects vary across cohorts or over event time, these comparisons can generate biased aggregate estimates.

This concern applies directly here. Agromercados were introduced in staggered cohorts between July 2024 and May 2025, and treatment effects may differ across opening dates and post-treatment horizons. A pooled TWFE regression would therefore risk conflating cohort-specific responses with the overall average effect.

To address this, the baseline specification uses the [Callaway and Sant’Anna \(2021\)](#) (C&S 2021) doubly robust estimator. This approach identifies a group-time average treatment effect, $ATT(g, t)$, for each cohort g and period t , using only not-yet-treated units as controls. These group-time effects are then aggregated into an overall ATT, avoiding the contamination problem inherent in TWFE under staggered timing.

The information channel requires a separate second step. Because $Info_t$ varies within the post-treatment period, including it directly in the first-stage DiD would risk introducing a bad-control problem ([Donald and Lang, 2007](#)). The analysis therefore proceeds sequentially: first, estimate the baseline $ATT(g, t)$ terms; second, regress these estimated effects on $Info_t$ using inverse-variance weighted least squares with cohort and event-time-bin fixed effects ([Luxen and Vetter, 2011](#)). Because $Info_t$ operates at the group-time level, a pooled regression risks

get rid of downward bias in standard errors due to common cohort (group) errors. By isolating the variance to the second-stage aggregated estimates, this two-step procedure corrects the asymptotic distribution of the t-statistic for small and finite group sizes, ensuring valid and conservative statistical inference for the information mechanism

Baseline TWFE estimates are reported only as a robustness check in Appendix C

Chapter 3

Institutional Framework

The Agromercados Programme

El Salvador’s retail food markets are served by a network of district markets with multiple small vendors, supplied by intermediaries. Retail competition is therefore decentralised: these vendors set prices independently, and price comparison across outlets is costly for consumers, particularly for fresh produce.

In response to food price increases driven by adverse weather and supply disruptions, MAG launched the agromercados programme in July 2024 as a direct price-stabilisation measure. The programme operates public retail outlets at which products are sold at a centrally set and publicly announced weekly price, P_t^{gov} , and disseminated through official channels and on-site displays. Private retail prices are monitored independently by the Defensoría del Consumidor, which collects weekly price data for a basket of 12 basic vegetables across 43 district markets. The staggered rollout of the programme and the subsequent information blackout in November 2024 provide the natural experiment that underpins the empirical analysis, I use in this study.

3.1 District Markets and Agromercados: A Structural Comparison

Understanding why the programme’s disciplining effect is partial rather than complete requires a clear picture of what the two market types look like in practice, and how fundamentally they differ in structure, pricing, and the information they make available to consumers.

District markets are the permanent, covered market halls located in each district seat and monitored by the Defensoría. Stall space is rented by private vendors who are predominantly intermediaries: they source produce from wholesale platforms and mark up prices before selling to end consumers. Prices at each stall are set independently; there is no public price posting, no price ceiling, and no obligation to match a competitor. Payment is overwhelmingly cash-based, and vendors commonly extend informal credit to regular customers. Because fresh vegetables

are sold loose by weight or count, without barcodes or standardised packaging, direct price comparison across stalls requires a physical visit to each one. It is this combination of many vendors, independent pricing, no posted prices, and cash-and-credit relationships that sustains the search frictions and equilibrium dispersion described in Section 2.1.

Agromercados differ on every one of these dimensions. They are state-operated retail outlets that purchase directly from registered producer cooperatives, eliminating intermediaries. All products within a given agromercado are sold at a weekly price, P_t^{gov} , which is uniform across all sites and publicly announced in advance. There is no price negotiation, no credit, and no vendor heterogeneity within the outlet: every tomato is the same price at every agromercado on a given week. The physical format, typically an open-air or semi-covered setup staffed by cooperative representatives, is also distinct from the enclosed, multi-vendor market halls of the district system. Table 3.1 summarises the key structural differences.

Table 3.1: District markets versus agromercados: structural comparison

Dimension	District market	Agromercado
Operator	Private vendors	MAG / state
Supply chain	Wholesale intermediary	Direct from producer cooperative
Pricing	Vendor-set, independent, unposted	Centralised P_t^{gov} , publicly announced
Price transparency	Low (requires physical visit per stall)	High (broadcast weekly)
Price heterogeneity	High (within and across stalls)	None (uniform within outlet)
Payment	Cash, informal credit common	Cash, no credit
Schedule	Open daily from 6:00 AM–6:00 PM	Open daily from 6:00 AM–1:00 PM
Wait times	Many vendors, small wait times	Consumers must queue up and wait longer

Notes: This comparison is based on programme documentation, district market knowledge, and MAG press releases. District markets refer specifically to the 43 Defensoría-monitored district market locations in the analysis sample.

This structural contrast is not merely descriptive: it is the foundation for understanding the mechanism outcomes documented in the subsequent sections. The agromercado removes the intermediary mark-up and publishes P_t^{gov} , which theory predicts should force private prices downward. However, it does not remove the non-price advantages that keep regular customers loyal to their district-market vendor: established relationships, proximity, informal credit, and the convenience of an enclosed, all-weather market. These are the search and transport frictions from the theoretical framework. A public competitor, with substitutable products, can overcome them only if consumers know its price and find it low enough to justify the switch.

3.2 Agromercado Programme Roll-out and Price Signalling

Three features of programme design are central for identification. First, agromercados were rolled out gradually: an initial cohort of eight sites opened on 6 July 2024, with additional locations coming online through late July and subsequent months, reaching more than sixty

sites by early 2025. This staggered rollout generates cross-district variation in the date of first exposure that underpins the [Callaway and Sant’Anna \(2021\)](#) design.

Second, from programme launch through late November 2024, MAG publicly broadcast P_t^{gov} weekly via official social media channels, press releases, and on-site displays. This active information phase—denoted $\text{Info}_t = 1$ —is referred to throughout as the *Agro+Info* phase.

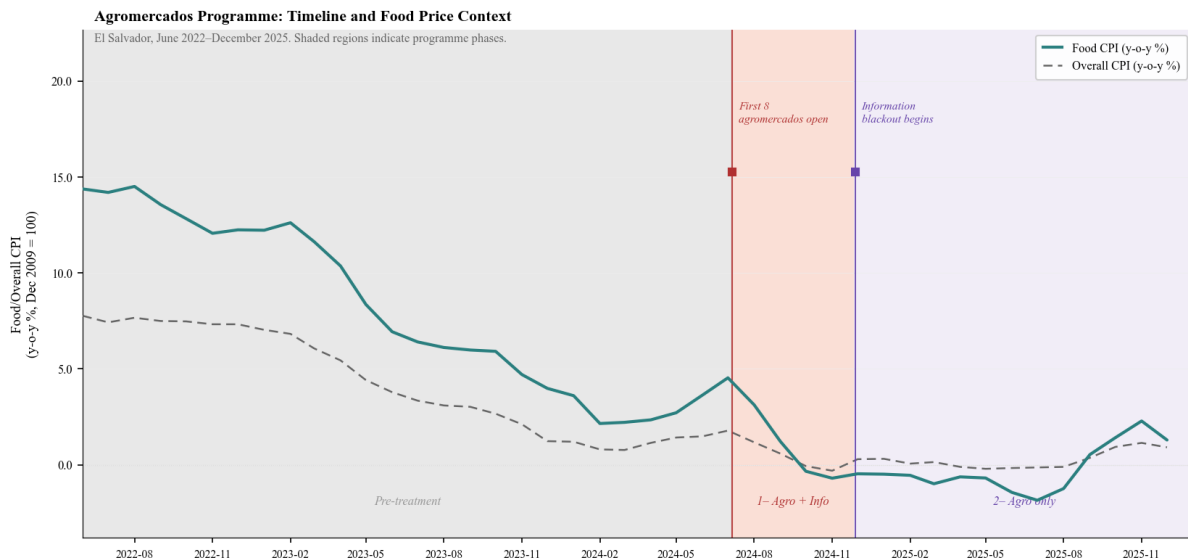


Figure 3.1: Agromercados Programme: Timeline and Food Price Context, El Salvador, June 2022–December 2025.

Notes: Shaded regions denote the pre-treatment window (light grey, June 2022–July 2024), Phase 1 in which agromercados operated alongside active national broadcasting of P_t^{gov} (pink, 6 July–28 November 2024), and Phase 2 in which physical markets remained open but price broadcasting ceased (light purple, December 2024 onwards). Vertical markers indicate two key programme milestones: first eight agromercados open (6 July 2024, solid red line); and information blackout begins (28 November 2024, solid purple line). The solid teal line plots the year-on-year percentage change in the food and non-alcoholic beverages sub-index of the *CPI*; the dashed grey line plots the overall *CPI* (base: December 2009 = 100). Source: [Banco Central de Reserva de El Salvador \(2026\)](#).

Third, on 28 November 2024, following public statements by the Central Bank of El Salvador attributing deflationary pressures in part to agromercado pricing, MAG ceased the national broadcast of P_t^{gov} . Physical markets continued to operate and new sites opened, but the information channel was switched off for the subsequent ten months ($\text{Info}_t = 0$). This administrative decision—driven by macroeconomic rather than local market considerations—constitutes the *information blackout* that enables identification of the information channel as a distinct causal force.

Figure 3.1 contextualises these three features jointly. The Food Consumer Price Index series for El Salvador documents the inflationary episode that motivated the programme, with year-on-year food price growth peaking above 14 per cent in mid-2022 before gradually declining through 2023. The July 2024 launch coincides with a transient price spike driven by adverse weather supply disruptions, precisely the shock the programme was designed to absorb.

Chapter 4

Data & Sample Construction

Study Overview

In this study, I measure the causal impact of the Agromercados programme on private markets by utilising two main data sources: (1) a high-frequency biweekly district–market–product panel spanning 43 Defensoría-monitored district markets, 12 canonical vegetables, and 95 bi-weekly periods (May 2022 – December 2025); and (2) an agromercado dataset constructed from open-source intelligence, which provides treatment timing and location data for the 61 agromercados and the source for the two main treatment indicators: *Agro*, which captures the physical presence of an agromercado within a 5 km road-network radius of a district market, and *Info*, which captures the national information signal during the price-broadcasting phase.

Parting from these two treatment indicators, I define two outcome tracks (variables) upon which both treatments are estimated.

Track 1: Price levels. The log mean unit price ($\ln P_{it}^{\text{priv}}$) at the district–product–period level (27,825 observations), testing the programme’s direct price-anchoring effect on private retail competition.

Track 2: Price dispersion. The within-district basket coefficient of variation CV_{it} at the district–period level (2,611 observations), testing whether agromercado entry reduced inter-period price volatility as a proxy for food security improvement.

A detailed description of the data sources, acquisition process, cleaning, and treatment construction is provided in Sections 4.1 to 4.4.

4.1 Data Sources and Acquisition

The empirical analysis relies on two main datasets: a panel of private district–market prices from the *Defensoría del Consumidor* (Consumers’ Defence Office; “Defensoría”) and a dis-

district-level dataset of agromercado locations, opening dates and official prices assembled from Ministry of Agriculture communications and open source material.

Weekly unit prices across district markets were obtained through a formal public data-access request submitted to the *Defensoría*. The raw *Sondeo de Precios* (Price Survey) files contain 1,496,187 price quotes for basic food products from 192 district markets in 14 departments between 6 January 2022 and 20 February 2026. These survey records constitute the primary source for all private-market outcome variables used in this thesis.

A parallel request was submitted to the MAG for agromercado opening dates, locations and weekly official prices. This request was denied; the agromercado dataset was therefore constructed using Open Source Intelligence (OSINT). Official MAG social media posts ([Ministerio de Agricultura y Ganadería, 2024](#)), press releases, and national news reports were collected to identify each site’s opening date, geographic coordinates, and the weekly public price vector P_t^{gov} . This made it possible to recover both the timing of the agromercado rollout and the public price signal. The direct-observation window for P_t^{gov} covers July to November 2024. Including some sporadically recovered observations from weeks in the post-blackout period, for missing observations a carry-forward imputation strategy is applied; results are reported separately for the observed sub-sample as a robustness check, and the imputed flag (`gov_imputed_any`) is used to define Info_t . Agromercado and district market coordinates are geocoded using GeoPython; road-network distances are computed via Open Source Routing Machine (OSRM), using OpenStreetMap data, capturing actual travel frictions more accurately than linear distance. ([Luxen and Vetter, 2011](#)) ([OpenStreetMap contributors, 2026](#))

4.2 Sample Construction

The raw *Defensoría* data required extensive cleaning and product name harmonisation before analysis. Product descriptions are not standardised across markets or over time, so a product crosswalk was used to align all *Defensoría* item names to standardised names, matching the initial 14 products announced by MAG at the programme launch. After cleaning, only products that could be matched consistently, across both datasets over time were kept. Thus, ensuring comparability between private and public price series. The final canonical basket consists of 12 staple vegetables. Observations with implausible unit prices, inconsistent units of measurement or severe missingness are removed during cleaning.

Weekly observations are aggregated to a biweekly frequency (14-day periods) to increase within-cell price observations and reduce noise from occasional missing survey weeks. A coverage rule is applied: a district market is retained if at least 40 per cent of all biweekly periods in the sample window are observed. This rule reduces the sample from 192 to 43 district markets with adequately dense price histories over the full window.

Two panel structures are constructed from the cleaned data. *Track 1* (price levels) has as its observation unit a district-product-biweekly-period cell, with the main outcome being the log

mean unit price of each product in each district and period (`log_price_mean`). *Track 2* (price dispersion) uses a district–period panel, with the outcome being the within-district coefficient of variation of the canonical basket (`basketCVit`), computed over all products in each biweekly period. Periods in which fewer than two price quotes are observed in a given district are dropped from the `CVit` calculation. Dispersion is calculated directly from observed prices rather than from imputed series. Table 4.1 summarises the final panel structure.

Table 4.1: Panel structure and coverage

Dimension	Value
Panel observations	
Track 1: District–product–period cells	27,825
Track 2: District–period cells	2,611
Geographic coverage	
Total district markets	43
Treated (≤ 5 km from agromercado)	32
Never-treated (> 5 km)	11
Time and product coverage	
Canonical products in basket	12
Bi-weekly periods	95
Panel start	23 May 2022
Panel end	29 December 2025
Period length	14 days
Coverage threshold	$\geq 40\%$ periods observed

Notes: Treatment status is defined under the 5 km road-network radius specification. Track 1 evaluates the absolute price level for individual products. Track 2 evaluates the basket coefficient of variation (CV); periods with fewer than two valid price quotes in a given district are omitted from the dispersion calculation.

4.3 Treatment and Control Group Assignment

Careful treatment assignment is the backbone of the empirical design, thus it warrants a detailed description. The treatment of interest is the presence of an operational agromercado within a certain radius of the district market, as well as the presence of the information signal. This section describes how treatment status is defined.

The unit of observation. The analysis is built on a panel of 43 Defensoría-monitored *district markets*, each identified by a unique geocoded location corresponding to the municipal market hall where the Defensoría’s enumerators collect weekly price quotes. These are the private-market data points. Separately, each of the 61 agromercados that opened between July 2024 and early 2025 also has a geocoded location. Treatment is then determined by the spatial relationship between these two sets of locations.

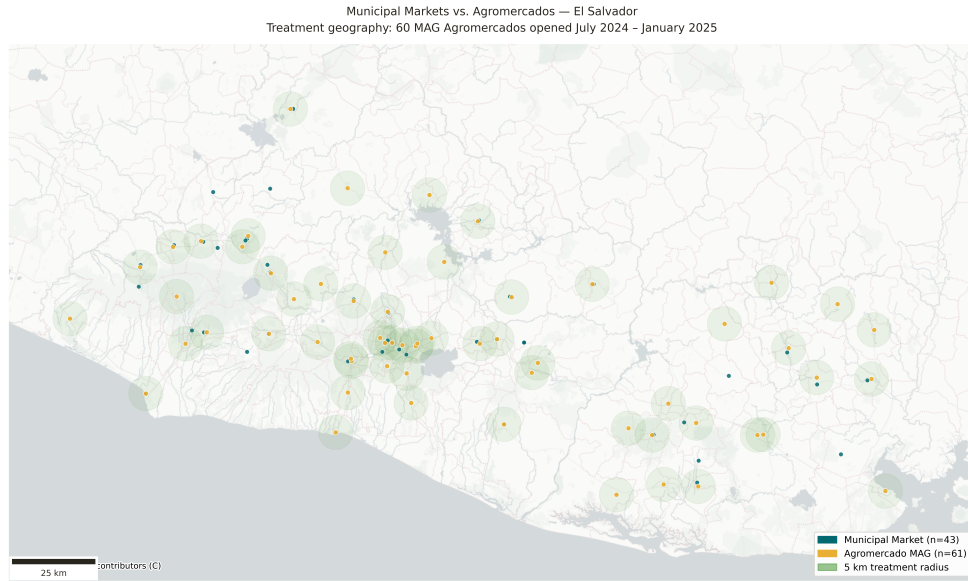


Figure 4.1: Agromercados and District Market– Treatment Assignment.

Notes: District markets and agromercados in El Salvador. Teal points show *Defensoría* district markets ($N = 43$); orange points show agromercados; shaded circles indicate the 5 km treatment radius used to define treated districts

Why road-network distance? The relevant measure of proximity is not straight-line (Euclidean) distance but road-network distance, computed via ORSM. This distinction matters for the following reason: the theoretical mechanism operates through the consumer’s decision of whether to travel to an agromercado rather than purchase at the local district market. That decision depends on actual travel time and cost, which are determined by roads, not straight lines. In El Salvador’s urban geography, a district market and an agromercado that are 3 km apart may be connected by a direct arterial road, or they may be separated by a river, a hill, or an absence of direct road connections, making the trip far more burdensome than the Euclidean distance suggests.

The 5 km treatment threshold. For each district market, the OSRM road-network distance to every agromercado that is operational in each biweekly period is computed. A district is classified as *treated* from the first biweekly period in which at least one agromercado lies within a 5 km road-network radius. The 5 km threshold is chosen on substantive grounds: at 5 km by road in El Salvador’s urban and peri-urban context, travel time by bus or motorcycle taxi is typically 10–20 minutes, which is a plausible outside option for regular grocery shoppers making multiple weekly trips. Consumers at greater distances face travel times and costs that are unlikely to make a substitution trip worthwhile for routine produce purchases.

Treated districts and treatment cohorts. Of the 43 retained districts, 32 lie within 5 km of at least one agromercado at some point during the sample window and constitute the *treated group*. Because agromercados opened on different dates, these 32 districts entered treatment at different times, forming 12 *treatment cohorts* (see Appendix A.1) defined by the biweekly

period of first exposure. The first cohort—those districts nearest to the eight agromercados that opened on 6 July 2024, began treatment earliest. Districts whose nearest agromercado opened in August, September, or subsequent months constitute later cohorts. This staggered structure creates heterogeneity that TWFE mishandles, but it also generates the within-programme variation in timing that the Callaway-Sant’Anna estimator exploits.

Control groups: not-yet-treated and never-treated. The remaining 11 districts, for which no agromercado opened within 5 km during the entire sample window, constitute the *never-treated* group. In principle, these are the cleanest controls. In practice, eleven districts is a small group, and the comparison is less representative.

Therefore, in this thesis, I use a more robust control group: the *not-yet-treated* districts. At each cohort-time pair (g, t) —that is, for each cohort g and calendar period t , the comparison group consists of all districts that have not yet been treated as of period t , including both the never-treated eleven and the later-cohort treated districts that have not yet received their first agromercado.

The information indicator. The treatment defined above captures agromercado presence. A second indicator, $Info_t \in \{0, 1\}$, is defined at the national level rather than the district level: it equals one during the Agro+Info phase (July–November 2024, when MAG actively broadcasts P_t^{gov}) and zero during the Agro-only phase (December 2024 onwards). This indicator is the same for all districts at any given point in time. In the estimating equations, it is interacted with $Agro_{it}$ to ask: among districts that have an agromercado nearby, do prices differ between periods when P_t^{gov} is being broadcast and periods when it is not?

4.4 Descriptive Statistics

Table 4.2 reports summary statistics for the main outcome variables, stratified by treatment status under the 5 km radius specification. The average private market price across the full district–product–period panel is \$0.63 per unit (SD = \$0.59). The log-price difference between treated and never-treated districts at baseline is -0.015 ($t = -1.22$, $p = 0.22$), which is not statistically significant, supporting pre-intervention price-level balance in Section ?? and Figures 4.2.

Price dispersion, measured by the basket coefficient of variation (CV) within each district and period, averages 0.212 across the full sample. Treated districts exhibit higher baseline dispersion ($CV_{it} = 0.218$) relative to never-treated districts ($CV_{it} = 0.192$), a gap of 0.025 points (13.2 % of the never-treated mean; $t = 13.66^{***}$). This is consistent with treated districts being located in denser urban or peri-urban markets where heterogeneous vendors and higher search costs sustain wider price variation—and motivates the use of price dispersion as a secondary outcome in the empirical analysis. Crucially, this pre-treatment dispersion gap does not

threaten identification: the Callaway & Sant’Anna (2021) estimator conditions on district fixed effects, absorbing time-invariant differences in baseline dispersion levels.

Table 4.2: Summary statistics by treatment status (5 km radius specification)

Variable	Sample	<i>N</i>	Mean	SD	p10	Median	p90	Diff.	<i>t</i> -stat
<i>Panel A: Price level (USD per unit)</i>									
Full sample		27,825	0.630	0.594	0.163	0.333	1.550		
Treated (≤ 5 km)		21,513	0.626	0.589	0.165	0.333	1.525	−0.018	−2.02**
Never-treated		6,312	0.644	0.608	0.156	0.338	1.610		
<i>Panel B: Log price (Track 1)</i>									
Full sample		27,825	−0.836	0.847	−1.812	−1.099	0.438		
Treated (≤ 5 km)		21,513	−0.839	0.842	−1.801	−1.099	0.422	−0.015	−1.22
Never-treated		6,312	−0.824	0.862	−1.856	−1.084	0.476		
<i>Panel C: Price coefficient of variation (Track 2)</i>									
Full sample		27,133	0.212	0.141	0.075	0.197	0.341		
Treated (≤ 5 km)		20,958	0.218	0.146	0.078	0.202	0.348	+0.025	13.66***
Never-treated		6,175	0.192	0.122	0.066	0.182	0.306		

Notes: Unit of observation is a district–product–biweekly period cell. Treated districts lie within 5 km of at least one operational agromercado; never-treated districts have no agromercado within 5 km during the full sample window (May 2022 – December 2025). *Diff.* is the treated minus never-treated mean difference, reported on the treated row. *t*-statistics test equality of means (Welch two-sample *t*-test). Price CV is the within-district coefficient of variation of the canonical 12-product basket. ** $p < 0.05$, *** $p < 0.01$.

Figure 4.2 plots the raw bi-weekly mean price level and basket CV for treated and never-treated districts over the full sample window. Rather than relying solely on the formal econometric tests in Chapter 6 event study plots, this raw data provides immediate visual support for the underlying parallel-trends assumption.

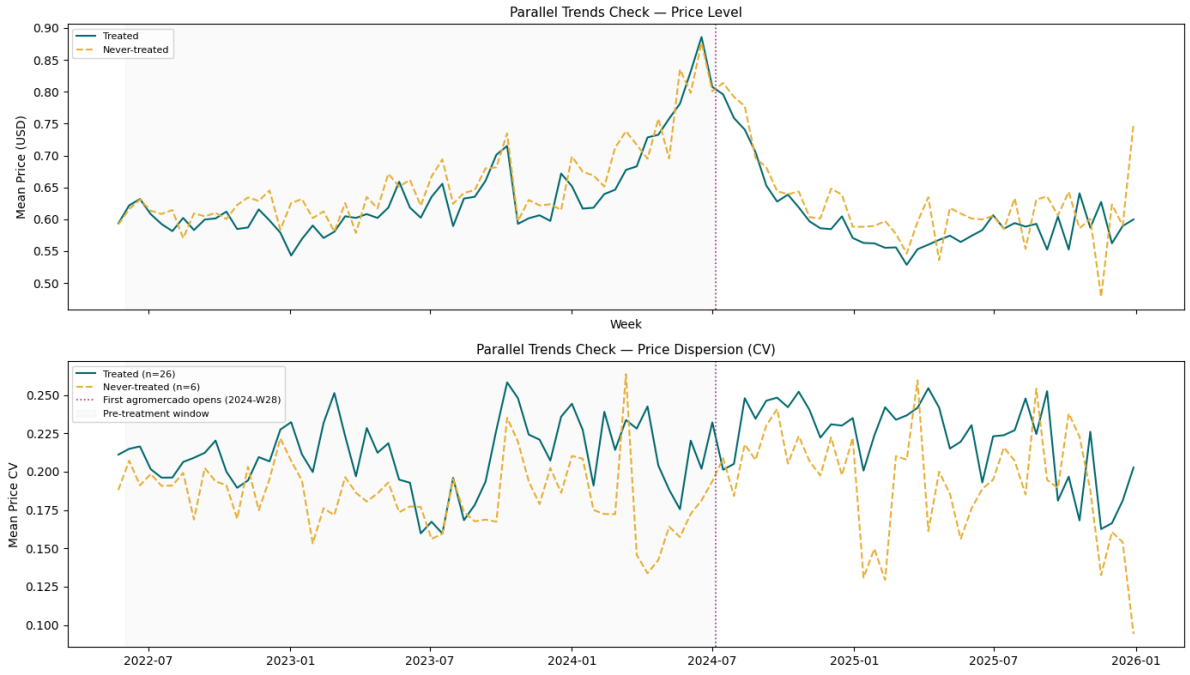


Figure 4.2: Raw Time Series of Market Outcomes and Visual Evidence of Parallel Trends

Notes: The panels plot the unadjusted bi-weekly mean outcomes for the treated cohort (districts located within 5 km of at least one operational agromercado during the sample window; $N = 32$) and the never-treated group (districts with no agromercado within 5 km; $N = 11$). Panel A displays the mean unit price level (USD); Panel B displays the within-district basket coefficient of variation (CV). The vertical dotted line denotes the first agromercado opening in July 2024, marking the beginning of the staggered treatment rollout. Series are aggregated at the district–biweekly-period level.

As the panels demonstrate, prior to the initial policy intervention in July 2024, the trajectories of both cohorts co-move closely. In Panel A, the price levels for both groups respond almost identically to aggregate macroeconomic shocks, tracking each other tightly through the supply-driven inflation spike of mid-2024. In Panel B, while treated districts exhibit a structurally higher baseline dispersion—consistent with their location in denser urban hubs with higher search frictions. Thus, their pre-treatment temporal variations closely track those of the never-treated group. This stable pre-treatment co-movement indicates at a first glance that the never-treated districts serve as a valid counterfactual for unobserved macroeconomic pricing dynamics.

Chapter 5

Empirical Strategy

5.1 Identification

The empirical design exploits two sources of quasi-experimental variation. First, agromercados were rolled out gradually across districts, generating variation in the timing of first exposure to a nearby public outlet. Second, on 28 November 2024, MAG halted the national broadcasting of the weekly public price vector while physical agromercados continued to operate. This blackout provides separate variation in the information channel.

The main treatment indicator, Agro_{it} , switches on when district i first lies within the treatment radius of an operational agromercado. The information indicator, Info_t , varies only at the calendar-time level and equals one during the broadcasting phase and zero during the blackout phase. Identification therefore proceeds in two steps: first, estimating the average effect of nearby agromercado entry; second, testing whether those estimated effects differ systematically between broadcast and blackout periods.

Throughout, not-yet-treated districts serve as the primary comparison group. Pre-treatment event-time coefficients are used to assess the parallel-trends assumption.

5.2 Baseline treatment effects

The baseline analysis estimates the effect of nearby agromercado entry on two outcomes. Track 1 studies the **anchoring effect**, using the log mean unit price of product p in district i and bi-weekly period t , denoted $\ln P_{ipt}^{\text{priv}}$. Track 2 studies **price stability as a proxy for food security**, using the within-district basket coefficient of variation, CV_{it} , computed over the harmonised 12-product basket.

The main estimand is the group-time average treatment effect on the treated, following [Callaway and Sant’Anna \(2021\)](#):

$$\text{ATT}(g, t) = \mathbb{E}[Y_{it}(g) - Y_{it}(\infty) \mid G_i = g], \quad (5.1)$$

where $Y_{it}(g)$ denotes the potential outcome for a unit first treated in cohort g , and $Y_{it}(\infty)$ denotes the untreated counterfactual. The overall average treatment effect is obtained by aggregating the post-treatment (g, t) cells, while the pre-treatment cells are used to assess parallel trends in event time.

For exposition, the corresponding baseline specification in levels can be written separately for each outcome track as:

$$\ln P_{ipt}^{priv} = \alpha + \beta_0 Agro_{it} + \delta_i + \theta_t + \varepsilon_{ipt}, \quad (5.2)$$

$$CV_{it} = \alpha + \beta_0 Agro_{it} + \delta_i + \theta_t + \varepsilon_{it}, \quad (5.3)$$

where $Agro_{it}$ switches on when district i first lies within the treatment radius of an operational agromercado, δ_i are district fixed effects, and θ_t are bi-weekly period fixed effects. In the empirical analysis, these baseline effects are estimated using the doubly robust Callaway–Sant’Anna staggered DiD estimator, with not-yet-treated districts as the comparison group.

5.3 Information signalling

The information channel is estimated in a second step using the group-time treatment effects recovered from the baseline Callaway–Sant’Anna specification. Let $\widehat{ATT}^{\ln P}(g, t)$ denote the estimated treatment effect for Track 1 on log prices, and $\widehat{ATT}^{CV}(g, t)$ the corresponding estimate for Track 2 on the basket coefficient of variation.

The second-stage specification is:

$$\widehat{ATT}(g, t) = \alpha + \beta \text{Info}_t + \lambda_g + \eta_{b(e)} + u_{gt}, \quad e = t - g, \quad (5.4)$$

where Info_t equals one during the price-broadcasting phase and zero during the blackout phase, λ_g are cohort fixed effects, and $\eta_{b(e)}$ are event-time-bin fixed effects. The event time $e = 0$ is excluded so that the treatment-onset period is not confounded with the subsequent treatment path.

The coefficient β captures whether the estimated treatment effects differ systematically between periods with and without active public price broadcasting, conditional on cohort and event time. Under the information-signalling hypothesis, a negative β implies that broadcasting strengthens the treatment effect: lower private prices in Track 1 and lower within-district price dispersion in Track 2.

5.4 Spatial Decay and the Radius of Relief

The mixed-oligopoly framework predicts that agromercados discipline private pricing only within the range where consumers can plausibly substitute. Following Hotelling (1929), the constraint on a private vendor at road-network distance d_i from the nearest agromercado is:

$$P_{it}^{priv} \leq P_t^{gov} + \tau d_i, \quad (5.5)$$

where τ is the per-kilometre transport cost faced by consumers. Put plainly: the further a consumer lives from the nearest agromercado, the more a private vendor can charge before losing that consumer to the public outlet. Similarly, the farther away a district market is from agromercado, the more rapid the disciplining effect decays. The empirical question is therefore whether the estimated treatment effect on private prices diminishes as d_i grows.

Distance-bin specification. To test this, I split the treated sample into three concentric bands: 0–1 km, 1–3 km, and 3–5 km from the nearest agromercado; and estimate a separate post-treatment effect for each band relative to never-treated districts:

$$\ln P_{ipt}^{priv} = \alpha + \sum_{b \in \mathcal{B}} \beta_b \left(\mathbf{1}\{d_i \in b\} \times \text{Post}_{it} \right) + \delta_{ip} + \theta_t + \varepsilon_{ipt}, \quad (5.6)$$

where $\mathcal{B} = \{[0, 1), [1, 3), [3, 5)\}$ km, Post_{it} switches on from the first bi-weekly period of agromercado exposure, δ_{ip} are district-product fixed effects, and θ_t are bi-weekly period fixed effects. Standard errors are clustered at the district level. If the Hotelling mechanism holds, the $\hat{\beta}_b$ sequence should be most negative in the 0–1 km band and attenuate towards zero as distance increases.

Bin-interacted event study. The baseline distance-bin specification yields a single post-treatment average per spatial band. To examine the dynamic timing of these effects and formally assess the parallel-trends assumption, Equation (5.6) is extended to an event-study framework.

Let $e_{it} = t - g_i$ denote event time, tracking the number of bi-weekly periods relative to a district's first exposure to an agromercado. To preserve statistical power and smooth high-frequency variation, these periods are aggregated into six relative-time bins, indexed by s : four pre-treatment and two post-treatment. The dynamic model is estimated as:

$$\ln P_{ipt}^{priv} = \alpha + \sum_{b \in \mathcal{B}} \sum_{s \neq s_{\text{ref}}} \gamma_{bs} \left(\mathbf{1}\{d_i \in b\} \times \mathbf{1}\{e_{it} \in s\} \right) + \delta_{ip} + \theta_t + \varepsilon_{ipt}, \quad (5.7)$$

where $\mathcal{B} = \{[0, 1), [1, 3), [3, 5)\}$ is the set of spatial distance bands in kilometres, and s_{ref} denotes the omitted reference category (the earliest pre-treatment bin, $\text{Pre} \leq -6$). Omitting this

bin anchors the estimates to a stable baseline long before the intervention. As in the baseline specification, δ_{ip} are district-product fixed effects, and θ_t are bi-weekly period fixed effects.

The parameters of interest, γ_{bs} , measure the average price deviation for districts in distance band b during event-time bin s , relative to the omitted pre-treatment baseline. The intuition is straightforward: if the pre-treatment coefficients ($\hat{\gamma}_{bs}$ for $s < 0$) are statistically indistinguishable from zero, this provides empirical support for the parallel-trends assumption. The post-treatment coefficients ($\hat{\gamma}_{bs}$ for $s \geq 0$) subsequently map the temporal and spatial emergence of the competitive radius of relief.

5.5 Mechanism Tests

The baseline and spatial estimates show whether agromercados affected private prices on average and whether any effect varied with distance. They do not, however, reveal whether private vendors actually moved towards the public benchmark price. The purpose of the mechanism tests is therefore interpretive: to assess whether agromercados operated as a binding outside option or only as an incomplete reference price.

In this section, I use two tests. The first examines whether the private–public price gap narrows or widens with continued exposure to agromercados. The second examines whether changes in the official public price are passed through symmetrically into private prices.

5.5.1 Private–public price gap

The baseline ATT measures the average effect of agromercado entry on private prices. It does not reveal whether vendors gradually adjusted towards the public benchmark or whether the two price series drifted further apart as the programme matured. This distinction matters: if the gap narrows with exposure, the agromercado is functioning as a tightening anchor; if it widens, vendors are decoupling from the public price despite continued proximity to the outlet, which is consistent with residual market power rather than genuine competitive discipline.

To test this directly, I define the log price gap as:

$$\Delta_{ipt} = \ln P_{ipt}^{priv} - \ln P_{pt}^{gov}, \quad (5.8)$$

where P_{pt}^{gov} is the official agromercado price for product p in period t . Within the post-treatment sample of treated districts, I then estimate:

$$\Delta_{ipt} = \alpha + \kappa Exposure_{it} + \delta_i + \delta_p + \gamma_g + \varepsilon_{ipt}, \quad (5.9)$$

where $Exposure_{it}$ counts the number of bi-weekly periods since district i first gained access to a nearby agromercado, and γ_g are treatment-cohort fixed effects. The coefficient κ is the parameter of interest. A negative κ implies that the private–public price gap closes the longer

a district is exposed to agromercado competition. A positive κ implies the opposite: vendors progressively widen the mark-up above the public benchmark, meaning the agromercado fails to become a binding constraint despite continued operation.

5.5.2 Asymmetric pass-through

Even if private vendors do not converge fully to P_{pt}^{gov} , the public benchmark may still matter if changes in the official price are transmitted into private-market prices. I test this using changes in logs, so the coefficients can be read as pass-through elasticities: they measure the percentage change in private prices associated with a 1 percent change in the public price.

$$\Delta \ln P_{ipt}^{priv} = \beta^+ \Delta \ln P_{pt}^{gov} \mathbf{1}[\Delta \ln P_{pt}^{gov} > 0] + \beta^- \Delta \ln P_{pt}^{gov} \mathbf{1}[\Delta \ln P_{pt}^{gov} < 0] + \delta_i + \delta_p + \varepsilon_{ipt}, \quad (5.10)$$

estimated over the window in which official weekly prices are directly observed. Here, β^+ captures pass-through from public price increases, while β^- captures pass-through from public price decreases. Under a genuinely binding public anchor, private vendors should respond in broadly symmetric fashion, so upward and downward changes in P_{pt}^{gov} should generate similar responses in absolute value. By contrast, $\beta^+ > |\beta^-|$ implies that vendors track public price increases but do not pass through public price decreases, which is consistent with residual local market power and incomplete anchoring.

5.6 Theoretical Scorecard for Results Interpretation

Before presenting the full interpretation of the estimates, Table 5.1 maps out the theoretical and empirical patterns expected under different mechanisms. The purpose of this scorecard is not to identify the treatment effect by itself, but to guide the results interpretation: the same baseline ATT may be consistent with different underlying theory and mechanisms, however, when it is read jointly with the information interaction and the spatial decay test, a more complete picture emerges.

As such, this heuristic tool, outlines four different scenarios, each with a distinct pattern across the three tests. The first two scenarios represent more ideal cases of full convergence and information-driven convergence, while the last two represent different versions of partial convergence under residual frictions. The actual results are interpreted in light of this scorecard in Chapter 6.

Table 5.1: Identification scorecard: predicted joint patterns by theoretical scenario

Scenario	Baseline ATT	Info tion	interac- tion	Spatial gradi- ent	Interpretation
1. Full convergence	< 0 , signifi- cant	≈ 0		≈ 0	Physical competition alone disciplines private sellers. (Obradovits and Plaickner, 2023)
2. Info-driven convergence	≈ 0	< 0 , signifi- cant		≈ 0	The public price signal shifts consumers towards price-directed search. (Ding and Zhang, 2018)
3. Frictions persist	≈ 0	≈ 0 or > 0		≈ 0	Search and transport frictions fully insulate local vendors. (Hotelling, 1929)
4. Partial convergence	weakly nega- tive or ≈ 0	< 0 , signifi- cant		> 0 , signifi- cant	The public competitor constrains prices locally, but competitive discipline decays with distance.

Notes: The relevant mechanism is inferred from the joint pattern across all three estimates rather than from any single coefficient in isolation.

Chapter 6

Results

6.1 Price Level and Dispersion

Table 6.1 reports the aggregated Callaway–Sant’Anna (C&S) estimates for the two main outcomes: log private-market prices (Track 1) and within-district basket price dispersion, measured by the coefficient of variation (CV, Track 2).

Table 6.1: Effect of Agromercados on Prices and Price Dispersion — Overall ATT

	(1) Log Price Level ($\ln P_{it}^{priv}$)	(2) Basket CV (CV_{itit})
Overall ATT	−0.0216 (0.0564)	0.0004 (0.0158)
Mean pre-treatment ATT	0.0003	−0.0003
Mean post-treatment ATT	−0.0324	0.0085
Pre-trend violation ($p < 0.10$)	Yes	Yes
Observations (N)	27,825	2,611
Units (N)	516	43
Unit definition	district–product	district
Periods	95	95
Control group	Not-yet-treated districts	
Estimator	Doubly Robust (C&S 2021)	
District FE		✓
Period FE		✓

Notes: Overall average treatment effect on the treated (ATT) from the doubly-robust [Callaway and Sant’Anna \(2021\)](#) staggered difference-in-differences estimator. Control group: not-yet-treated districts at each cohort–period pair (g, t) . Standard errors computed via the influence-function variance estimator, clustered at the district level. Pre-trend violation indicates at least one pre-treatment $\widehat{ATT}(g, t)$ cell significant at the 10% level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The central result is unambiguous. For log prices, the overall average treatment effect on the treated (ATT) is -0.0216 ($SE = 0.0564$, $p = 0.702$); for basket CV it is $+0.0004$ ($SE = 0.0158$, $p = 0.982$). Both estimates are statistically indistinguishable from zero.

Nevertheless, an empirically estimated null is itself informative. Physical entry of the state as a subsidised retail competitor was not, by itself, sufficient to generate market-wide price convergence in surrounding private markets. The price estimate is not only statistically insignificant but also economically small relative to sampling uncertainty; the dispersion estimate is effectively centred around zero. Together, these magnitudes suggest that any competitive effect of agromercado entry was either too weak, too localised, or too heterogeneous to register as a clear district-level average treatment effect.

The event-study plots in Figures 6.1 and 6.2 display the full dynamic profile over the first year of treatment. Post-treatment coefficients remain close to zero for both outcomes, with no systematic downward drift.

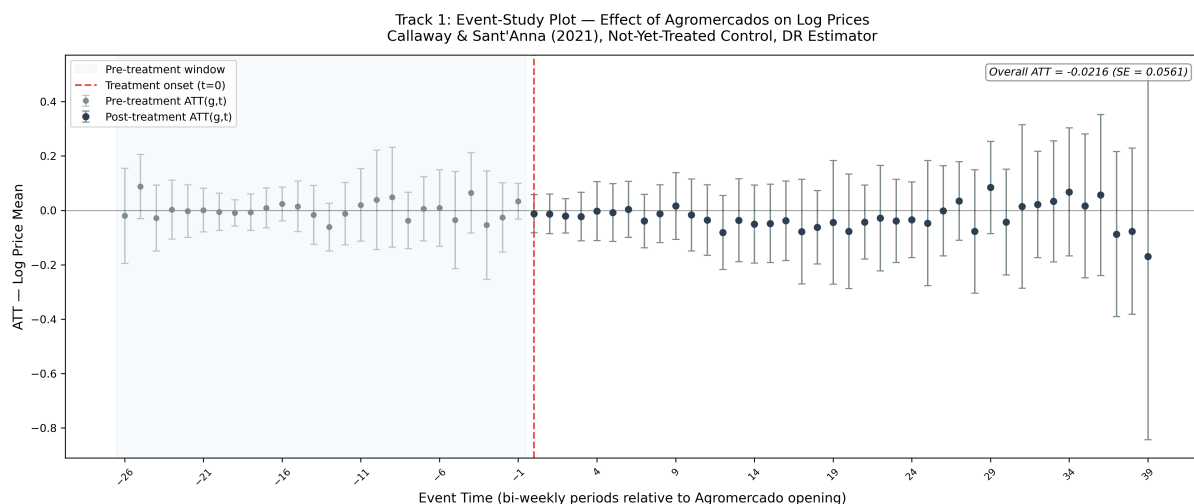


Figure 6.1: Event-Study Plot — Effect of Agromercados on Log Vegetable Prices (Track 1).
Notes: Event-study estimates from the doubly-robust [Callaway and Sant’Anna \(2021\)](#) staggered difference-in-differences estimator. Outcome: log mean unit price of product p in district i at biweekly period t . Control group: not-yet-treated at each cohort-period pair (g, t) . Event time measured in biweekly periods relative to first agromercado opening within 5 km ($t = 0$, red dashed line). Grey points: pre-treatment $ATT(g, t)$ cells; dark points: post-treatment cells. Error bars: 95% confidence intervals (doubly-robust influence-function variance). Standard errors clustered at the district level.

Pre-treatment dynamics differ across outcomes. For basket CV, pre-treatment estimates are broadly flat, which is broadly consistent with the parallel-trends assumption, though some significant pre-trend cells are present and should be noted. For log prices, pre-treatment coefficients display more movement prior to treatment, consistent with administrative targeting of districts experiencing sharper price pressures before agromercado entry. Price-level estimates should therefore be interpreted with greater caution than dispersion estimates throughout.

The post-treatment evidence provides little support for the view that physical agromercado entry alone imposed sufficient competitive discipline to lower private prices or compress basket dispersion on average. This motivates the mechanism tests in Sections 6.2 through 6.4: rather

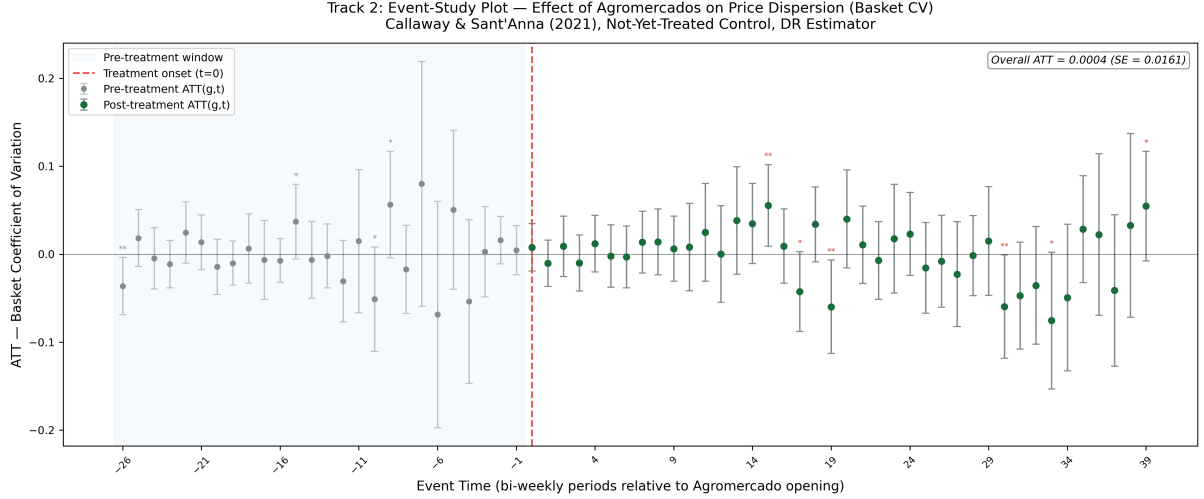


Figure 6.2: Event-Study Plot — Effect of Agromercados on Price Dispersion, Basket CV (Track 2).

Notes: As for Figure 6.1, except the outcome is the within-district basket CV of the canonical 12-product basket at biweekly period t . Periods with fewer than two price quotes per district are excluded. All other estimation details identical to Track 1.

than asking only whether the policy worked on average, the following sections examine whether effects were conditional on information, exposure time, and road-network distance.

6.2 The Information Channel

Table 6.2 reports the second-stage meta-regression estimates testing whether active public price broadcasting ($\text{Info}_t = 1$) impacted the post-treatment $\widehat{ATT}(g,t)$ price and coefficient of variation estimates recovered from the baseline C&S 2021 baseline regressions. All nine treatment cohorts are retained. Identification of $\hat{\beta}$ comes from variation in Info_t across post-treatment cohort–period cells, as the broadcasting phase and the subsequent blackout both occur within the calendar span covered by the treated sample.

Identification is supported by the pre-treatment falsification test. When equation (5.4) is re-estimated on the 567 pre-treatment cells, the placebo coefficient is statistically indistinguishable from zero for both log prices ($\hat{\beta}_{\text{placebo}} = -0.0113$, $p = 0.541$) and basket dispersion ($\hat{\beta}_{\text{placebo}} = 0.0097$, $p = 0.445$).

Turning to the main results, the information coefficient is not statistically significant in either track. For log prices, $\hat{\beta} = 0.0225$ (SE = 0.0150, $p = 0.134$), indicating that treatment effects are slightly less negative during the broadcasting phase. For basket dispersion, $\hat{\beta} = -0.0202$ (SE = 0.0137, $p = 0.139$), suggesting marginally lower dispersion when broadcasting is active. The 95% confidence intervals are $[-0.0069, 0.0519]$ for log prices and $[-0.0471, 0.0066]$ for basket CV, ruling out large effects in either direction.

Taken together, these results provide no evidence that active price broadcasting independently shifted average treatment effects in either direction. This null is not attributable to a

Table 6.2: Two-Step Sequential Meta-Regression: Effect of Information Broadcasting on $\widehat{ATT}(g, t)$

	(1) Log Price Level ($\ln P_{it}^{priv}$)	(2) Basket CV (CV_{itit})
$\hat{\beta}$: $\text{Info}_t = 1$	0.0225	−0.0202
SE	(0.0150)	(0.0137)
95% CI	[−0.0069, 0.0519]	[−0.0471, 0.0066]
p -value	0.1336	0.1391
Pre-treatment placebo p -value	0.5410	0.4453
N ($ATT(g, t)$ cells)	214	214
Cohorts retained	9	9
$\text{Info}_t=1$ coverage (post)	22.6%	22.6%
R^2	0.3280	0.3678
Weights	$1/\hat{\sigma}^2$	$1/\hat{\sigma}^2$
Standard errors	HC3	HC3
Cohort FE (λ_g)	✓	✓
Event-time bin FE ($\eta_{b(e)}$, 8-week bins)	✓	✓
$e = 0$ excluded	✓	✓
Cohorts dropped	None	None

Notes: Dependent variable is $\widehat{ATT}(g, t)$ from the doubly robust [Callaway and Sant’Anna \(2021\)](#) estimator, restricted to post-treatment cells ($e > 0$). $\text{Info}_t = 1$ during the active price-broadcasting phase (prior to 28 November 2024), and 0 during the administrative suspension phase. Observations are weighted by inverse estimated variance ($1/\hat{\sigma}_{g,t}^2$). Event-time bins have width of 4 biweekly periods (8 weeks). The pre-treatment placebo re-estimates the same specification on $e < 0$ cells only ($N = 567$); 50 pre-treatment (g, t) cells record $\text{Info}_t = 1$ but do not enter the primary regression. Because the number of cohorts is too small for reliable cluster-robust inference, HC3 heteroskedasticity-robust standard errors are reported throughout. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

pre-existing trend, as the falsification test confirms. The evidence is consistent with a market in which neither the physical presence of the agromercado nor the price signal alone was sufficient to produce a statistically detectable shift in $\widehat{ATT}(g, t)$ at the cohort–period level. The mechanism tests in Section 6.3 examine instead whether the *structure* of private price responses to the public benchmark – i.e., the direction and symmetry of pass-through, is informative about how vendors used the state price anchor.

6.3 Pass-Through and the Price Gap

Table 6.3 examines why agromercados did not generate market-wide price discipline even when the public benchmark was visible. Panel A tests whether the private–public price gap narrows or widens with exposure time. Panel B tests whether changes in the official price are transmitted symmetrically to private prices. In this context, *pass-through* denotes the extent to which a change in P_t^{gov} is transmitted into private-market prices.

Panel A shows that the gap widens rather than narrows. The coefficient on relative exposure time is positive and significant in both the full post-treatment sample (0.0176, $p < 0.01$) and the observed-price sub-sample (0.0673, $p < 0.01$). Private vendors progressively move *away* from the public benchmark, which is inconsistent with a tightening anchor. Full specification robustness across four estimators is reported in Table A.4 (Appendix A).

Panel B turns to pass-through. The symmetric specification yields an average pass-through of 0.2705: only around 27% of any change in the public price is transmitted to private prices on average, which already signals incomplete discipline. The asymmetric decomposition reveals a stark contrast. Positive shocks are passed through strongly ($\hat{\beta}^+ = 0.4813$, $p < 0.001$). Negative shocks are not transmitted to private prices in any statistically detectable way ($\hat{\beta}^- = -0.0618$, SE, =, 0.0510, $p = 0.226$). Private vendors thus respond to upward movements in the public benchmark but show no corresponding adjustment when P_t^{gov} falls.

In practice, a 1% increase in the public price is associated with a 0.48% increase in private prices, while a 1% decrease in P_t^{gov} produces only a 0.06% fall. Together, Panels A and B point to the same underlying structure: private vendors selectively use the public benchmark as a *ceiling reference* rather than a binding constraint, adjusting upward when P^{gov} rises — which raises the threshold at which consumers consider switching — but resisting downward movements that would require compressing their own mark-ups (Ding and Zhang, 2018). This pattern is inconsistent with full price-directed search and instead characteristic of residual local market power sustained by captive consumers, informal credit relationships, and location specificity. By-product pass-through coefficients, which show substantial heterogeneity across the 12-product basket, are reported in Panel B of Table A.4 (Appendix A).

Table 6.3: Price Gap Trend and Pass-Through Mechanism (Post-Treatment Sample)

Panel A: Price gap trend $\Delta_{ipt} = \ln P_{ipt}^{priv} - \ln P_{pt}^{gov}$		
	(1) Full post-treatment	(2) Observed prices only
Relative exposure time	0.0176*** (0.0051)	0.0673*** (0.0081)
Observations (N)	7,104	2,306
R^2 (within)	0.061	0.155
Entity FE	✓	✓
Month FE	✓	✓
Panel B: Pass-through of government price shocks $\Delta \ln P_{ipt}^{priv}$ on $\Delta \ln P_{pt}^{gov}$		
	(1) Symmetric	(2) Asymmetric
$\hat{\beta}$: $\Delta \ln P^{gov}$	0.2705*** (0.0404)	
$\hat{\beta}^+$: Positive shock ($\Delta \ln P^{gov} > 0$)		0.4813*** (0.0389)
$\hat{\beta}^-$: Negative shock ($\Delta \ln P^{gov} < 0$)		-0.0618 (0.0510)
Observations (N)	8,323	8,323
Product FE	✓	✓
District FE	✓	✓

Notes: Panel A estimates the price-gap trend on the post-treatment sample via panel OLS with entity and month fixed effects (Specification 2). Dependent variable is the log private–public price gap $\Delta_{ipt} = \ln P_{ipt}^{priv} - \ln P_{pt}^{gov}$. Column (2) restricts to non-imputed government price observations and periods prior to 30 November 2024. Panel B estimates pass-through of first-differenced government prices onto private prices across all treated-district observations. $\hat{\beta}^+$ and $\hat{\beta}^-$ capture transmission of positive and negative shocks respectively; a Wald test rejects $\hat{\beta}^+ = |\hat{\beta}^-|$ ($p < 0.01$). Note that $\hat{\beta}^-$ is not individually significant ($p = 0.226$), indicating that downward pass-through does not differ statistically from zero. Standard errors clustered at the district level throughout. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.4 Spatial Decay and the Radius of Relief

The theoretical framework predicts that the disciplining effect of an agromercado should weaken with road-network distance as transport and search costs restore local market power. To test this, I use a two-stage approach: first, a TWFE specification with concentric distance bins as the preferred estimator; second, an event-study by distance band to assess whether the spatial gradient is stable over time.

Distance-Bin TWFE

The preferred specification interacts an agromercado-active indicator with three concentric distance bins — 0–1 km, 1–3 km, and 3–5 km — relative to a not-yet-treated control group (Section 5.4, equation (5.6)). Under the 5 km treatment boundary, Table 6.4 reports none of the three bin coefficients reach the 10% significance threshold. The point estimates are small and negative for the 0–1 km and 3–5 km bins (-0.0125 and -0.0119 respectively) and negligible for the 1–3 km bin (-0.0005). No single radius of empirical relief is identified under this specification.

Event-Study by Distance Band

Figure 6.3 plots the event-study treatment effects disaggregated by distance bin (0–1 km, 1–3 km, 3–5 km) relative to the pre-treatment reference period. Pre-treatment coefficients are broadly flat across all three bands, which is consistent with the identifying assumption that assignment to distance bins is uncorrelated with pre-existing price trends. Post-treatment coefficients are uniformly small and imprecisely estimated, with no band showing a systematic negative trajectory. The 3–5 km bin records a statistically significant negative coefficient in the first post-treatment window (-0.0550 , $p < 0.001$), but this effect dissipates in the second window (-0.0046 , $p = 0.629$), suggesting it does not reflect a sustained disciplining effect.

Taken together, the spatial results are consistent with, but do not strongly corroborate, the theoretical prediction of a finite radius of relief. No single threshold distance can be identified at which competitive discipline sharply attenuates. This is likely due to two compounding limitations: the 5 km assignment radius may be too wide to generate meaningful variation in distance exposure within the treated sample, and the post-treatment window is insufficient to estimate a stable spatial gradient with precision. A tighter assignment threshold of 3 km, used as a robustness specification, shows a marginally significant spatial gradient for log prices and a stronger near/far differential for basket CV, consistent with the prediction that spatial decay operates within approximately 1–2 km of agromercado locations. These robustness results are reported in Appendix A.

Table 6.4: Spatial Decay: Distance-Bin TWFE Estimates (Track 1, Log Price Level)

	(1) Collapsed post-treatment	(2) Event-study bins (pre + post windows)
<i>Panel A: Collapsed post-treatment (reference: not-yet-treated)</i>		
$\hat{\gamma}_{[0-1 \text{ km}]}$	-0.0125 (0.0090)	
$\hat{\gamma}_{[1-3 \text{ km}]}$	-0.0005 (0.0088)	
$\hat{\gamma}_{[3-5 \text{ km}]}$	-0.0119 (0.0078)	
<i>Panel B: Event-study bins (reference: $Pre \leq -6$)</i>		
$\hat{\gamma}_{[0-1 \text{ km}]} \times Post_{[0,+4]}$		-0.0117 (0.0147)
$\hat{\gamma}_{[0-1 \text{ km}]} \times Post_{\geq 5}$		-0.0115 (0.0103)
$\hat{\gamma}_{[1-3 \text{ km}]} \times Post_{[0,+4]}$		0.0074 (0.0122)
$\hat{\gamma}_{[1-3 \text{ km}]} \times Post_{\geq 5}$		-0.0015 (0.0114)
$\hat{\gamma}_{[3-5 \text{ km}]} \times Post_{[0,+4]}$		-0.0550*** (0.0145)
$\hat{\gamma}_{[3-5 \text{ km}]} \times Post_{\geq 5}$		-0.0046 (0.0094)
Observations (N)	25,459	25,459
Units (N)	456	456
Entity FE	✓	✓
Period FE	✓	✓
SE	Clustered at district level	

Notes: Column (1) estimates equation (5.6) and Column (2) estimates equation (5.7), both via panel OLS with entity and time fixed effects. Distance bins are defined by OSRM road-network distance from the district centroid to the nearest operational agromercado. Reference group: not-yet-treated districts at each period. Column (1) uses a single collapsed post-treatment indicator interacted with each distance bin. Column (2) uses the full event-study parameterisation with pre- and post-treatment windows: $Pre \in [-5, -4]$, $Pre \in [-3, -2]$, $Pre \in [-1]$, $Post [0, +4]$, $Post [\geq 5]$; reference period $Pre \leq -6$. Five districts with fewer than eight post-treatment periods are excluded. The only statistically significant coefficient is $\hat{\gamma}_{[3-5 \text{ km}]} \times Post_{[0,+4]}$ ($p < 0.001$), which dissipates in the subsequent window ($p = 0.629$), indicating a transient rather than sustained effect. No empirical radius of relief is identified under the 5 km boundary. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

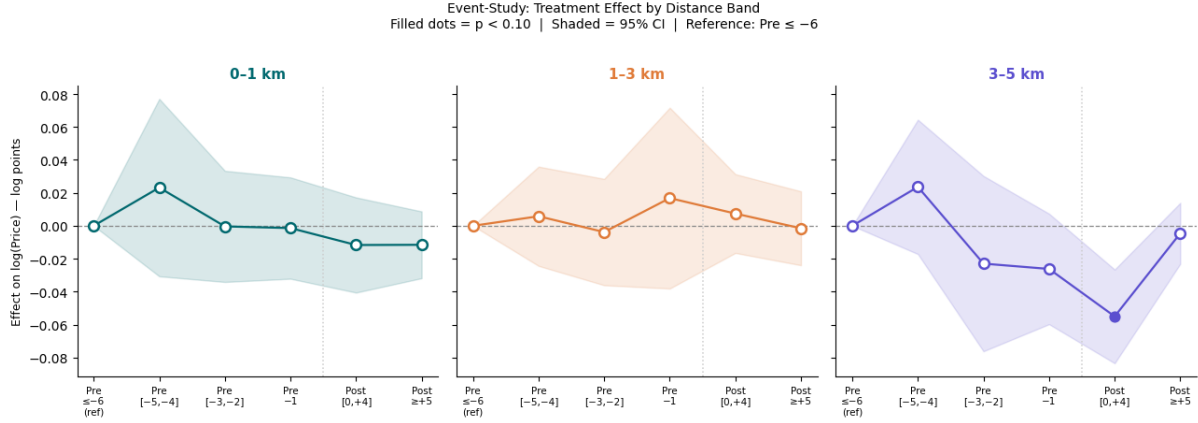


Figure 6.3: Event-Study Treatment Effect by Distance Band (Track 1).

Notes: TWFE event-study with distance-bin $\times$ relative-period interaction dummies. Three distance bins: 0–1 km (teal), 1–3 km (orange), 3–5 km (purple). Reference period: $\tau \leq -6$ (omitted). Pre-treatment windows: $\tau \in \{-4, -3\}$ (Pre 2), $\tau \in \{-2, -1\}$ (Pre 1). Post-treatment windows: $\tau \in \{0, \dots, 4\}$ (Post 1), $\tau \geq 5$ (Post 2). Filled dots: $p < 0.10$; open dots: $p \geq 0.10$. Shaded band: 95% CI. Entity and period fixed effects included. Standard errors clustered at the district level. Road-network distances computed via OSRM.

6.5 Scorecard Verdict

Returning to Table 5.1, the joint pattern of results is most consistent with Scenario 4 (partial convergence under residual frictions). The baseline ATT is a precisely estimated null. The information meta-regression is underpowered and inconclusive, precluding a causal statement on the information channel, though the mechanism evidence remains informative. The price gap widens with exposure time; pass-through is sharply asymmetric ($\hat{\beta}^+ = 0.481^{***}$ vs. $\hat{\beta}^- = -0.062$, insignificant); a Wald test formally rejects symmetry ($p < 0.01$); and the spatial distance-bin specification identifies no clear radius of competitive discipline under the 5 km boundary, with marginal evidence of short-range effects emerging only in the tighter 3 km robustness design.

Together, these findings indicate that the agromercado programme operated as a *localised and incomplete price anchor*. Physical presence was necessary but not sufficient; the information signal could not be cleanly separated from treatment onset given the cohort structure; and structural frictions — captive consumers, informal credit, and location specificity — prevented the state price from functioning as a binding outside option for most private vendors. The state improved price coordination more than it imposed price discipline.

Chapter 7

Conclusion

7.1 Summary of Findings

In this study, I sought to determine whether El Salvador's Agromercados programme successfully reduced private food prices and within-district price dispersion in nearby district markets, and to identify the mechanisms driving these outcomes. By utilising a doubly robust staggered difference-in-differences estimator, the central finding is a precisely estimated aggregate null. At the district level, physical agromercado entry did not produce a statistically detectable reduction in average private prices or basket price dispersion.

In the theoretical scorecard developed in Chapter 5, this pattern is most consistent with partial convergence under residual frictions, scenario 4. In other words, the programme introduced a salient public benchmark, but did not transform that benchmark into a sufficiently strong outside option to induce market-wide repricing by private vendors. The overall null average effect therefore does not imply irrelevance. Rather, it indicates that any competitive pressure generated by state entry was too weak, too localised, or too heterogeneous to emerge as a clear district-level average treatment effect.

The mechanism evidence helps untangle this explain why. First, the private–public price gap widened with exposure time rather than narrowing. This is inconsistent with a market in which continued agromercado presence progressively tightened private pricing around the public benchmark. Instead, private vendors appear to have moved away from the state price as the programme matured. Second, pass-through was sharply asymmetric. Private vendors responded to upward movements in the public benchmark, but not to downward ones. That pattern is difficult to reconcile with a fully binding public price anchor. A more plausible interpretation is that the state price functioned as a reference point or ceiling signal, while incumbent vendors preserved their mark-ups whenever downward adjustment would have been required. In that sense, the programme constrained pricing narratives more than it constrained pricing power. Third, the results on the information channel remain inconclusive. The second-stage meta-regression indicates no statistically significant effect of active public price broadcasting

on either private price levels or basket dispersion. Although the point estimate for dispersion is directionally consistent with mild compression under broadcasting, the evidence is not strong enough to support a causal claim. Given the overlap between treatment timing and the broadcasting phase, that part of the analysis is best interpreted as descriptive rather than definitive.

The spatial evidence points in the same direction. Under the preferred 5 km treatment boundary, no stable radius of competitive discipline is identified. Estimated distance-bin effects are small, and the only significant post-treatment effect is transient rather than sustained. The tighter 3 km robustness design provides only marginal support for the idea that any disciplining effect is highly local. Taken together, these findings suggest that transport costs, limited mobility, and the convenience advantages of incumbent vendors quickly restore local market power.

The study therefore answers the central research question in the following way. Agromercados did not reduce private food prices or price dispersion on average in surrounding district markets. The physical presence of a subsidised public seller was not sufficient to discipline private vendors unconditionally. Nor is there clean evidence that the national broadcasting of P_t^{gov} independently shifted outcomes. What the programme did achieve was more limited: it created a visible public benchmark that shaped private pricing behaviour at the margin, but did not overturn the structural frictions governing everyday retail competition. The state improved price coordination more than it imposed price discipline.

7.2 Policy Implications

Three policy implications follow from these findings.

Physical entry alone should not be expected to generate market-wide price discipline. A subsidised public outlet can create a visible low-price benchmark without forcing surrounding vendors to reprice. In this setting, non-price dimensions of competition remained intact: proximity, informal credit, established buyer–seller relationships, and shopping convenience continued to anchor consumers to local vendors. A policy that targets intermediary power only through state retail entry is therefore unlikely to deliver broad price convergence unless those frictions are also weakened.

Information is most valuable when consumers can realistically act on it. The logic of the programme treated public price broadcasting as a mechanism for reducing search frictions. That logic remains economically sound, even though the present estimates cannot isolate a clean causal information effect. If consumers are informed of the public benchmark but still face meaningful travel or switching costs, information alone will have limited force. The effectiveness of broadcasting therefore depends on whether the public option is practically reachable, not merely visible.

Outlet density matters more than nominal geographic reach. The spatial results imply that competitive pressure decays rapidly with road-network distance. For routine produce purchases, a broad coverage radius is a weak proxy for actual substitution possibilities. If the policy objective is to discipline private prices, a denser network of outlets in areas where consumers can switch at low cost is likely to be more effective than extending nominal coverage across wider distances. In practical terms, the location of outlets may matter at least as much as their number.

7.3 Limitations and Further Research

The conclusions of this thesis should be read subject to three main limitations. First, pre-treatment movement in the price-level estimates is consistent with some degree of administrative targeting, so the evidence on Track 1 requires more caution than the evidence on basket dispersion. Second, the information channel cannot be cleanly separated from treatment timing for most cohorts, which limits the strength of any causal interpretation of the broadcasting results. Third, the number of never-treated districts is small, which constrains precision in cohort-specific comparisons.

There are also data and horizon limitations. Direct observation of official government prices P_t^{gov} is concentrated in the broadcasting phase, which forces reliance on imputation later in the sample. In addition, the post-treatment window remains short relative to the behavioural adjustments implied by the theory. Consumer learning, switching behaviour, and persistent vendor repricing may all unfold over a longer horizon than the present data allow.

These limitations point to a clear agenda for future research. In my estimation, following evaluations would be especially valuable if national price broadcasting resumes, since that would allow the information channel to be studied with cleaner time variation. Alternatively, granular vendor level data on price and quantities sold, would successfully elevate the theoretical and empirical framework of my study. Allowing to draw more accurate empirical conclusions on the policy's effect on cartel formation, consumer and vendor welfare calculations. As well as the full quantitative impact of the policy, not just on competing P_{it}^{priv} prices, but also on amounts sold. Moreover, consumer-level data on shopping location, price awareness, and credit relationships would also allow for a more direct test of the search-friction mechanism than can be achieved with reduced-form price outcomes alone.

Broadly, the main lesson drawn from this study is conditional rather than universal. State retail entry can matter, but its effectiveness depends on the strength of the outside option it creates for consumers. Where transport costs, search frictions, and relational contracting remain strong, public sellers may operate as visible price islands rather than market-wide disciplinarian firms.

Bibliography

- Banco Central de Reserva de El Salvador. índice de precios al consumidor (IPC). Statistical series, Banco Central de Reserva de El Salvador, 2026. URL <http://estadisticas.bcr.gob.sv/serie/indice-de-precios-al-consumidor-ipc>. Accessed: May 2026.
- BBC News Mundo. Bukele en El Salvador: la apuesta del gobierno para bajar el alto precio de los alimentos (y las incógnitas que genera), October 2024. URL <https://www.bbc.com/mundo/articles/cq5e23nvx0yo>.
- Marc F. Bellemare. Rising food prices, food price volatility, and social unrest. *American Journal of Agricultural Economics*, 97(1):1–21, 2015. doi: 10.1093/ajae/aau038.
- Brantly Callaway and Pedro H. C. Sant’Anna. Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230, 2021. doi: 10.1016/j.jeconom.2020.12.001.
- Diario El Mundo. BCR atribuye la deflación a agromercados y central de abasto, December 2024. URL <https://diario.elmundo.sv/economia/bcr-atribuye-la-deflacion-a-agromercados-y-central-de-abasto>. Section: Economía.
- Min Ding and Jianan Zhang. Price-directed consumer search. *Management Science*, 64(1): 2025–2047, 2018. doi: 10.1287/mnsc.2016.2676.
- Stephen G Donald and Kevin Lang. Inference with difference-in-differences and other panel data. *The Review of Economics and Statistics*, 89(2):221–233, 2007.
- elsalvador.com. MAG recibe refuerzo de \$9 millones para logística de agromercados. *elsalvador.com*, 2024. URL <https://www.elsalvador.com/h-noticias/h-nacional/agricultura-ministerio-de-y-ganaderia-mercados/1187309/2024/>.
- Andrew Goodman-Bacon. Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277, 2021. doi: 10.1016/j.jeconom.2021.03.014.
- Harold Hotelling. Stability in competition. *The Economic Journal*, 39(153):41–57, 1929. doi: 10.2307/2224214.

- Legislative Assembly of El Salvador. Salvadoreños seguirán comprando alimentos a bajo costo en agromercados, 2025. URL <https://www.asamblea.gob.sv/node/13627>.
- Dennis Luxen and Christian Vetter. Real-time routing with OpenStreetMap data. In *Proceedings of the 19th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems*, pages 513–516. ACM, 2011.
- William C. Merrill and Norman Schneider. Government firms in oligopoly industries: A short-run analysis. *Quarterly Journal of Economics*, 80(3):400–412, 1966. doi: 10.2307/1880727.
- Ministerio de Agricultura y Ganadería. Official agromercados price broadcasts and rollout announcements [facebook page]. [https://www.facebook.com/\[Exact_MAG_Page_Name\]/](https://www.facebook.com/[Exact_MAG_Page_Name]/), 2024. Accessed: June 2026.
- Martin Obradovits and Philipp Plaickner. Price-directed search and competition. *RAND Journal of Economics*, 54(3):491–523, 2023. doi: 10.1111/1756-2171.12439.
- OpenStreetMap contributors. OpenStreetMap data. <https://www.openstreetmap.org>, 2026. Accessed: June 2026.
- Steven Salop and Joseph E. Stiglitz. Bargains and ripoffs: A model of monopolistically competitive price dispersion. *Review of Economic Studies*, 44(3):493–510, 1977. doi: 10.2307/2296903.
- Dale O. Stahl. Oligopolistic pricing with sequential consumer search. *American Economic Review*, 79(4):700–712, 1989.
- George J. Stigler. The economics of information. *Journal of Political Economy*, 69(3):213–225, 1961. doi: 10.1086/258464.
- Hal R. Varian. A model of sales. *American Economic Review*, 70(4):651–659, 1980.

Appendix A

Supplementary Results

This appendix reports supplementary material for the main 5 km specification used in Chapter 6. The tables below provide supporting descriptive evidence, cohort construction, and robustness checks that complement the main estimates without repeating the full discussion from the results chapter. Appendix B presents the alternative 3 km specification.

A.1 Agromercado Programme Cohort Construction

Table A.1 shows the staggered adoption structure underlying the Callaway–Sant’Anna design. Each cohort is defined by the first bi-weekly period in which the nearest agromercado became operational within 5 km of the district market. The cohort pattern confirms substantial variation in treatment timing, while the never-treated group remains small and is therefore used mainly as a clean comparison benchmark in the main analysis.

Table A.1: Staggered treatment cohort structure under the 5 km radius specification

Cohort	<i>N</i> districts	Mean dist. (km)
10 Jul 2024	8	1.19
11 Jul 2024	1	0.41
25 Jul 2024	2	1.21
31 Jul 2024	1	4.75
01 Aug 2024	4	1.68
13 Aug 2024	3	2.62
22 Aug 2024	4	1.12
19 Sep 2024	4	1.12
08 Nov 2024	2	1.79
10 Feb 2025	1	1.75
15 May 2025	1	0.60
21 May 2025	1	0.64
Never-treated ($G = 0$)	11	13.29

Notes: Cohorts are defined by the first bi-weekly period in which the nearest operational agromercado lies within 5 km road-network distance of the district market. Never-treated districts have no agromercado within 5 km during the full sample window and serve as the comparison group for the not-yet-treated design in Chapter 5.

A.2 Product-Level Summary Statistics

Table A.2 summarises the 12-product basket used throughout the thesis. The table shows substantial variation in both price levels and dispersion across vegetables, which is why price-dispersion results in Chapter 6 are informative at the basket level rather than for any single product. Tomatoes and several onion varieties display particularly high relative variation, while cabbage and broccoli are among the highest-priced items.

Table A.2: Product-level descriptive statistics

Product	N	Mean price	SD	Mean CV	p10	p90
Cabbage	2,492	2.002	0.449	0.186	1.544	2.571
Broccoli	2,469	1.374	0.238	0.187	1.083	1.679
Lettuce	2,535	0.957	0.180	0.209	0.750	1.179
Potato	2,516	0.679	0.162	0.109	0.500	0.894
Güisquil	2,259	0.396	0.151	0.213	0.250	0.559
Cucumber	1,400	0.306	0.063	0.195	0.230	0.390
Red onion	2,502	0.301	0.062	0.202	0.233	0.379
Carrot	2,556	0.300	0.063	0.244	0.228	0.379
White onion	2,575	0.275	0.059	0.211	0.211	0.350
Yellow onion	696	0.263	0.097	0.184	0.184	0.350
Green bell pepper	2,555	0.199	0.052	0.218	0.141	0.264
Tomato	2,578	0.138	0.049	0.351	0.090	0.201

Notes: Prices are per-unit USD values normalised by quantity purchased. Mean CV is the average within-district coefficient of variation across bi-weekly periods. Products are sorted by mean price in descending order.

A.3 Baseline Regressions: C&S Dynamic Event Study Estimates

Table A.3 presents the cohort-specific dynamic estimates for the two main outcomes under the preferred 5 km definition. The estimates are consistent with the main results in Chapter 6: average price effects remain close to zero, and basket dispersion does not show a robust post-treatment decline. Pre-treatment movement is more visible for the price-level outcome than for dispersion, which is why the main text places greater weight on the basket-CV track for causal interpretation.

Table A.3: C&S 2021 $\widehat{ATT}(g, t)$ over time: price level and basket CV

t	Pre-treatment			Info window					Agro only			
	-3	-2	-1	0	+1	+2	+3	+4	+5	+12	+14	+16
Panel A: Log price level ($\ln P_{ipt}^{priv}$)												
Agro	0.0085	0.0642	0.0763	-0.0124	-0.0206	-0.0027	0.0034	-0.0120	-0.0169	-0.0345	-0.0773	0.0221
(SE)	(0.0715)	(0.0763)	(0.0629)	(0.0374)	(0.0303)	(0.0574)	(0.0472)	(0.0604)	(0.0679)	(0.0714)	(0.1212)	(0.1057)
Overall ATT	-0.0216 (SE = 0.0574, $p = 0.698$)											
Panel B: Basket CV												
Agro	-0.0685	-0.0539	0.0159	0.0078	0.0089	0.0120	-0.0031	0.0138	0.0082	0.0229	-0.0015	-0.0355
(SE)	(0.0662)	(0.0491)	(0.0134)	(0.0134)	(0.0187)	(0.0172)	(0.0172)	(0.0183)	(0.0255)	(0.0244)	(0.0255)	(0.0311)
Overall ATT	0.0004 (SE = 0.0161, $p = 0.983$)											

Notes: Doubly robust Callaway–Sant’Anna estimator with `control_group = notyettreated`. Monthly intervals correspond to 0, 1, 2, 3, 4, 5, 12, 14, 16 months. Standard errors clustered at the district level. The estimates reproduce the main null results reported in Chapter 6.

A.4 Mechanism Robustness Test: Price Gap Trend and By-Product Pass-Through

Table A.4 reports the main mechanism robustness checks underlying the interpretation in Chapter 6. The price-gap trend remains positive across all specifications, which is consistent with the main finding that private vendors did not converge towards the public benchmark. The by-product pass-through estimates also mirror the main result: transmission is strong for several products, but incomplete overall, and the pattern remains asymmetric in economic terms.

Table A.4: Price Gap Robustness: Full Specification Comparison and By-Product Pass-Through

Panel A: Price gap trend Dep. var.: Δ_{ipt}				
	(1) Cohort FE	(2) Entity + Month FE	(3) Observed only	(4) TWFE saturated
Relative exposure time	0.0022*** (0.0005)	0.0176*** (0.0051)	0.0673*** (0.0081)	0.0057*** (0.0008)
Observations (N)	7,104	7,104	2,306	7,104
R^2	0.512	0.061	0.155	0.632
Entity FE		✓	✓	✓
Month FE		✓	✓	
Cohort FE	✓			✓

Panel B: By-product symmetric pass-through Dep. var.: $\Delta \ln P_{ipt}^{priv}$				
Product	$\hat{\beta}$	SE	p -value	N
Papa	0.6241***	(0.0874)	0.000	740
Tomate	0.5956***	(0.1392)	0.000	750
Repollo	0.5605***	(0.2054)	0.006	734
Zanahoria	0.4460**	(0.1793)	0.013	746
Güisquil	0.4145***	(0.0805)	0.000	619
Chile verde	0.4004***	(0.0856)	0.000	751
Lechuga	0.3521***	(0.0684)	0.000	741
Brócoli	0.2765**	(0.1244)	0.026	745
Cebolla blanca	0.2431***	(0.0513)	0.000	752
Pepino	0.1113	(0.0762)	0.144	738
Cebolla morada	0.0201	(0.0468)	0.668	750
Cebolla amarilla	-0.0818	(0.0883)	0.354	257
<i>Pooled</i>	0.2705***	(0.0404)	0.000	8,323

Notes: Panel A compares four specifications for the log price-gap trend. Specification (2) is the preferred estimator reported in Chapter 6. Specification (3) restricts to non-imputed government price observations prior to 30 November 2024. Specification (4) uses a saturated TWFE with cohort and entity fixed effects. Panel B reports product-level symmetric pass-through coefficients from OLS with district fixed effects and HC1 standard errors. Standard errors in Panel A are clustered at the district level. The appendix results support the main interpretation of incomplete anchoring in Chapter 6.

Appendix B

Robustness: 3 km Treatment Radius

This appendix reports the main results under the tighter 3 km treatment radius. The specification follows the same empirical design as Chapter 6, but treatment is assigned when an operational agromercado lies within 3 km road-network distance of the district market centroid. The purpose of this appendix is to test whether the main findings in Chapter 6 are driven by a broader 5 km exposure rule or whether they persist when competitive exposure is defined more restrictively. The 3 km estimates should therefore be read as a robustness check, not a replacement for the preferred specification.

B.1 Baseline Treatment Effects

Table B.1 summarises the overall ATT estimates for log private prices and basket price dispersion under the 3 km radius. Relative to the main results in Table 6.1, the price-level estimate is slightly more negative and the dispersion estimate turns modestly negative, but neither coefficient is statistically significant. The same identification caution remains: the event-study plots below continue to show pre-treatment movement in Track 1, while Track 2 is flatter but still imprecisely estimated.

B.2 Event-Study Plots

Figures B.1 and B.2 report the dynamic effects under the 3 km specification. The qualitative pattern is the same as in Chapter 6. Track 1 continues to display some pre-treatment movement, so the price-level estimates should still be treated with caution. Track 2 remains comparatively flatter pre-treatment, and post-treatment coefficients are small and imprecisely estimated. The tighter radius therefore improves exposure precision, but it does not deliver a qualitatively different conclusion from the main results.

Table B.1: Effect of Agromercados on Prices and Price Dispersion — Overall ATT (3 km Radius)

	(1) Log Price Level ($\ln P_{it}^{\text{priv}}$)	(2) Basket CV (CV_{it})
Overall ATT	−0.0289 (0.0512)	−0.0044 (0.0139)
Mean pre-treatment ATT	−0.0001	−0.0008
Mean post-treatment ATT	−0.0351	0.0039
Pre-trend violation ($p < 0.10$)	Yes	Yes
Control group	Not-yet-treated districts	
Estimator	Doubly Robust (C&S 2021)	
District FE	✓	
Period FE	✓	

Notes: Overall ATT from the doubly-robust [Callaway and Sant’Anna \(2021\)](#) staggered DiD estimator. Treatment is defined as the presence of at least one operational agromercado within 3 km road-network distance of the district market centroid. Standard errors are computed using the influence-function variance estimator and clustered at the district level. The 3 km design is more restrictive than the preferred 5 km specification in Chapter 6, but it does not alter the main conclusion: the average treatment effect remains statistically indistinguishable from zero.

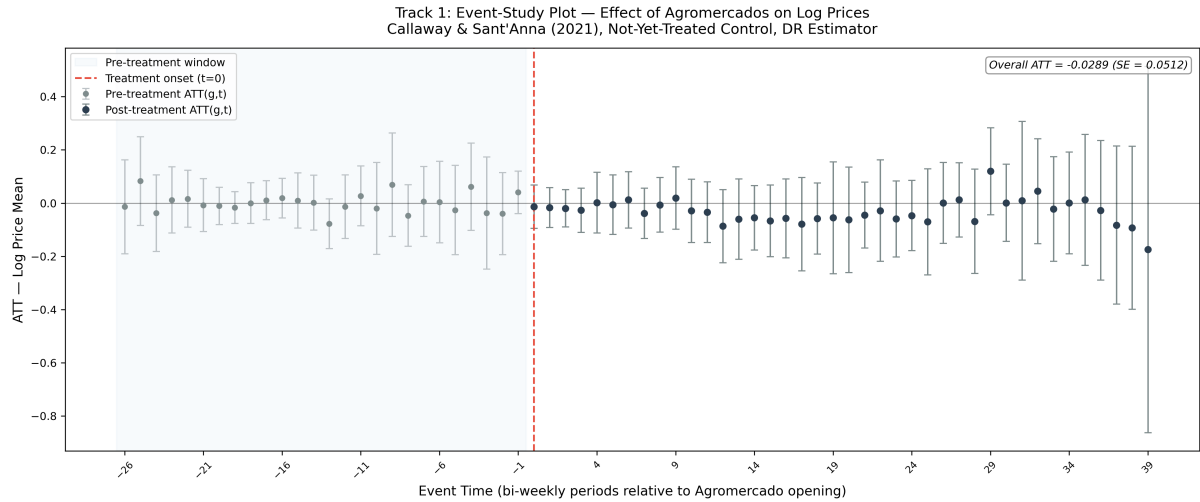


Figure B.1: Event-Study Plot (3 km) — Effect of Agromercados on Log Vegetable Prices (Track 1).

Notes: Event-study estimates from the doubly-robust [Callaway and Sant’Anna \(2021\)](#) staggered DiD estimator. The outcome is the log mean unit price of product p in district i at biweekly period t . The control group is the not-yet-treated set at each cohort-period pair (g, t) . Event time is measured relative to the first agromercado opening within 3 km ($t = 0$, red dashed line). Grey points denote pre-treatment $\widehat{ATT}(g, t)$ cells; dark points denote post-treatment cells. Error bars show 95% confidence intervals based on the doubly-robust influence-function variance estimator. Standard errors are clustered at the district level. The pre-treatment movement visible in this figure is the same issue noted in the main text.

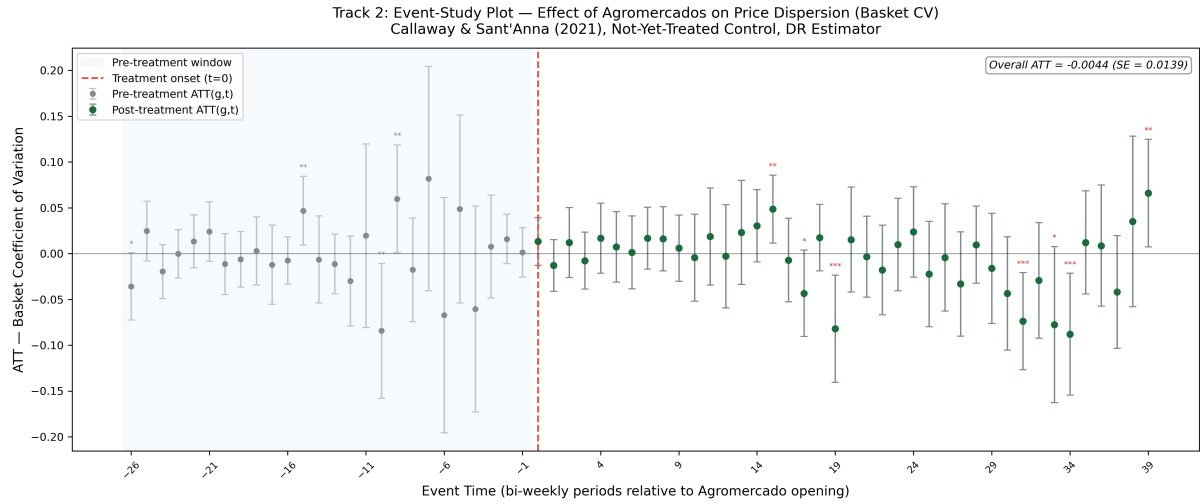


Figure B.2: Event-Study Plot (3 km) — Effect of Agromercados on Price Dispersion, Basket CV (Track 2).

Notes: As for Figure B.1, except the outcome is the within-district basket CV of the canonical 12-product basket at biweekly period t . Periods with fewer than two price quotes per district are excluded. All other estimation details are identical to Track 1. The dispersion path remains broadly flat before treatment and close to zero afterwards, which is consistent with the main specification in Chapter 6.

B.3 Price Gap and Pass-Through

Table B.2 replicates the mechanism tests from Section 6.3. The main conclusion is unchanged. The private–public price gap widens with relative exposure time, and the magnitude of the coefficient remains positive and statistically significant across all specifications. The preferred entity-and-month fixed-effects estimate is smaller than in the 5 km design, but it still points in the same direction. Likewise, pass-through remains asymmetric: positive government price shocks are transmitted to private prices, whereas negative shocks are not statistically distinguishable from zero. This closely matches the pattern in Chapter 6, where private vendors appear to treat the public price as a ceiling reference rather than a binding constraint.

B.4 Information Signalling Meta-Regression

Table B.3 reports the two-step meta-regression under the 3 km specification. The Track 1 information coefficient is positive and statistically significant at conventional levels, but the substantive interpretation is modest: private price effects are slightly less negative during the broadcasting phase than during the Agro-only phase. This is directionally consistent with the main chapter’s claim that broadcasting did not materially discipline private prices. For Track 2, the information coefficient remains statistically indistinguishable from zero. The placebos are far from significance, which is reassuring, but the same identification caveat remains: the channel is still best read as descriptive rather than cleanly causal.

Table B.2: Price Gap Trend and Pass-Through Mechanism — 3 km Radius

Panel A: Price gap trend $\Delta_{ipt} = \ln P_{ipt}^{priv} - \ln P_{pt}^{gov}$				
	(1) Cohort FE	(2) Entity + Month FE	(3) Observed only	(4) TWFE Saturated
Relative exposure time	0.0022*** (0.0006)	0.0122** (0.0052)	0.0555*** (0.0066)	0.0056*** (0.0009)
Observations (N)	6,069	6,069	1,996	6,069
R^2	0.508	0.062	0.148	0.624
Panel B: Pass-through of government price shocks $\Delta \ln P_{ipt}^{priv}$ on $\Delta \ln P_{pt}^{gov}$				
	(1) Symmetric		(2) Asymmetric	
$\hat{\beta}: \Delta \ln P^{gov}$	0.2584*** (0.0354)			
$\hat{\beta}^+$: Positive shock ($\Delta \ln P^{gov} > 0$)			0.4403*** (0.0353)	
$\hat{\beta}^-$: Negative shock ($\Delta \ln P^{gov} < 0$)			-0.0388 (0.0525)	
Observations (N)	7,184		7,184	

Notes: Panel A estimates the price-gap trend on the post-treatment sample. Column (2) is the preferred specification with entity month fixed effects. Column (3) restricts to non-imputed government price observations prior to 30 November 2024. Column (4) adds a full set of cohort $\times$ period interaction terms. Panel B estimates symmetric and asymmetric pass-through of first-differences in government prices onto private prices across all treated-district observations. A Wald test rejects symmetry ($p < 0.01$). Standard errors are clustered at the district level throughout. The 3 km estimates are directionally consistent with the main results in Chapter 4 but somewhat smaller in magnitude.

Table B.3: Two-Step Sequential Meta-Regression: Effect of Information Broadcasting on $\widehat{ATT}(g, t)$ — 3 km Radius

	(1) Log Price Level ($\ln P_{it}^{\text{priv}}$)	(2) Basket CV (CV_{itit})
$\hat{\beta}$: $\text{Info}_t = 1$	0.0357** (0.0174)	−0.0092 (0.0133)
95% CI	[0.0017, 0.0697]	[−0.0354, 0.0169]
Pre-treatment placebo $\hat{\beta}$ (p -value)	$p = 0.473$	$p = 0.420$
N ($ATT(g, t)$ cells)	207	207
R^2	0.316	0.203
Weights	$1/\hat{\sigma}^2$	$1/\hat{\sigma}^2$
Standard errors	HC3	HC3
Cohort FE (λ_g)	✓	✓
Event-time bin FE ($\eta_{b(e)}$, 8-week bins)	✓	✓
$e = 0$ excluded	✓	✓

Notes: Dependent variable is $\widehat{ATT}(g, t)$ from the doubly-robust [Callaway and Sant’Anna \(2021\)](#) estimator, restricted to post-treatment cells ($e > 0$). $\text{Info}_t = 1$ during the active price-broadcasting phase prior to 28 November 2024. Observations are weighted by inverse estimated variance ($1/\hat{\sigma}_{g,t}^2$). HC3 heteroskedasticity-robust standard errors are reported. The pre-treatment placebo re-estimates the same specification on $e < 0$ cells only. The positive Track 1 coefficient should be interpreted in the same restricted sense as in Chapter 6: it indicates attenuated competitive pressure during the broadcasting phase, not a robust causal effect of broadcasting on price levels.

Appendix C

TWFE Results Robustness

This appendix reports classical two-way fixed effects (TWFE) robustness checks for the main 5 km specification. They are included for comparison only: the preferred estimates in Chapter 6 use the Callaway–Sant’Anna staggered DiD design because treatment is staggered across cohorts and TWFE can mix treated units into the control group. The TWFE results are therefore best read as a conventional benchmark rather than the main causal estimate.

C.1 Track 1: Log Private Prices

Table C.1 reports TWFE estimates for log private prices under four specifications. The pattern is broadly consistent with the main results in Chapter 6: the agromercado indicator is small, negative in the Agro+Info specification, and statistically weak once the public price signal is controlled for in the observed-price sample. The information interaction is positive and statistically significant in the main Agro+Info specification, which is consistent with the interpretation that broadcasting mattered more as a signalling device than as a direct price-cutting channel. The imputed-price specification yields a positive public-price coefficient, but this should be interpreted cautiously because it relies on carry-forward construction of P_t^{gov} .

Table C.1: Effect of Agromercados on Log Private Prices ($\ln P_{ipt}^{priv}$)

	(1) Agro only	(2) Agro + Info	(3) Agro+Info+ P^{gov} obs.	(4) Agro+Info+ P^{gov} imputed
$\beta_1 : Agro_{it}$	-0.0063 (0.0000)	-0.0330*** (0.0000)	-0.0125 (0.0000)	0.0082 (0.0000)
$\beta_2 : Agro_{it} \times Info_t$		0.0420*** (0.0000)		-0.0197 (0.0000)
$\beta_3 : P_t^{gov}$ (control)				0.1029*** (0.0000)
District FE	✓	✓	✓	✓
Period FE	✓	✓	✓	✓
Control group mean	-0.8243	-0.8243	-0.8243	-0.8243
R^2 (within)	0.0030	0.0074	0.0120	0.0776
N	27,825	27,825	6,778	27,825

Notes: Classical TWFE estimations. Columns (1)–(2) are estimated over the full panel. Column (3) is restricted to the July–November 2024 window where P_t^{gov} is directly observed. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. These estimates are reported as a benchmark alongside the preferred staggered DiD results in Chapter 6.

C.2 Track 2: Basket Dispersion

Table C.2 reports the same TWFE benchmark for basket dispersion. The main conclusion is unchanged relative to Chapter 6: the agromercado programme does not generate a robust district-level decline in price dispersion. The Agro-only coefficient is near zero, the Agro+Info specification remains small, and the observed-price specification is only marginally positive. Taken together, these estimates do not overturn the main finding that dispersion effects are weak once staggered adoption is handled properly.

Table C.2: Effect of Agromercados on Basket Price Dispersion (CV_{it})

	(1) Agro only	(2) Agro + Info	(3) Agro+Info+ P^{gov} obs.	(4) Agro+Info+ P^{gov} imputed
$\beta_1 : Agro_{it}$	0.0006 (0.0000)	-0.0144 (0.0000)	0.0142** (0.0000)	-0.0045 (0.0000)
$\beta_2 : Agro_{it} \times Info_t$		0.0209 (0.0000)		0.0227 (0.0000)
District FE	✓	✓	✓	✓
Period FE	✓	✓	✓	✓
Control group mean	0.1912	0.1912	0.1912	0.1912
R^2 (within)	0.0006	0.0096	0.0039	0.0247
N	2,611	2,611	726	2,611

Notes: Outcome is the coefficient of variation of the 12-product canonical basket. Standard errors in parentheses, clustered at the district level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. These are robustness benchmarks; the preferred causal interpretation remains with the Callaway–Sant’Anna estimates in Chapter 6.